

Forecasting Water Levels in Hydropower Reservoirs

CS336 – Data Mining and Application

Group 10

Task Assignment Table

Student ID – Name	Task	Contribution
23521440 – Huynh Nhat Thanh	Collected hydropower data sources. Performed EDA, preprocessing, and feature engineering on both datasets (hydropower and weather). Prepared presentation slides	30%
23521441 – Le Duc Thanh	Crawled hydropower data. Conducted EDA, preprocessing, and feature engineering for hydropower data. Evaluated three models: CNN+LSTM, XGBoost, and Decision Tree. Developed the web demo with calculating the predicted results on those three models and EDA. Prepared presentation slides	30%
23521443 – Le Ngoc Thanh	Collected and crawled weather data. Performed preprocessing for weather data. Evaluated the CNN+LSTM model. Prepared presentation slides	25%
23521454 – Nguyen Tien Thanh	Collected and crawled weather data. Evaluated the XGBoost model. Developed the web demo with calculating the predicted results on XGBoost model. Prepared presentation slides	15%

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Introduction



Introduction

- **Forecasting hydropower reservoir water levels** is vital for efficient power generation, dam safety, and sustainable water management.
- It helps operators plan electricity production, prevent flooding, and ensure reliable energy supply.

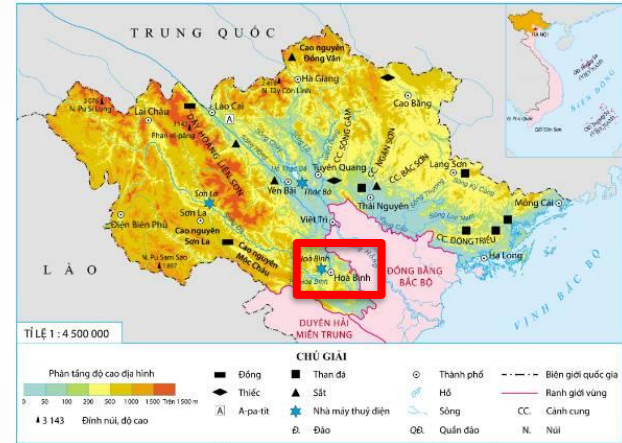
Introduction

- **The Hoa Binh Hydropower Plant**, located on the Da River in northern Vietnam, is one of the country's largest and most important hydroelectric facilities. Built with support from the former Soviet Union and completed in 1994, it has a total capacity of 1,920 MW. The reservoir plays a crucial role in electricity generation, flood control, and water regulation for the Red River Delta.

Introduction



Image of Hoa Binh Hydropower Dam



Hình 1. Lược đồ tự nhiên vùng Trung du và miền núi Bắc Bộ

Geographical Location

Introduction

- **The water level data of the Hoa Binh Hydropower Dam:** were collected from the official website of Vietnam Electricity (EVN) <https://hochuathuydien.evn.com.vn/PageHoChuaThuyDienEmbedEVN.aspx>



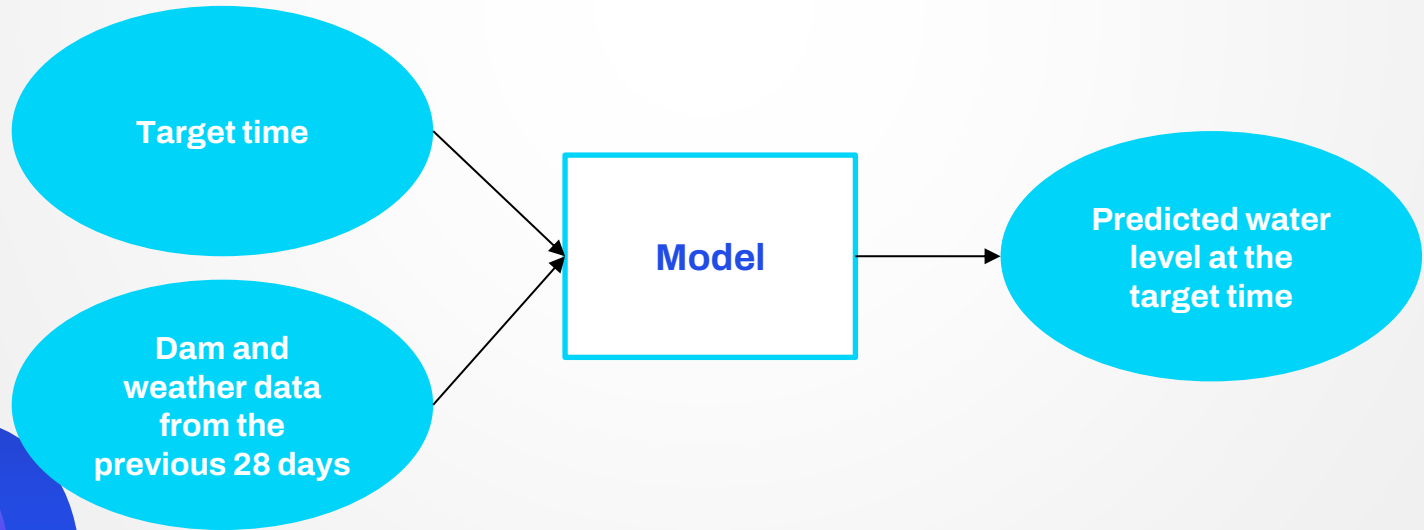
Introduction

- **The weather data:** were obtained from the Open-Meteo include variables such as temperature, precipitation, humidity, and wind speed, which are used to support hydropower reservoir forecasting <https://open-meteo.com/en/docs/historical-weather-api>



Introduction

- **Input:** **Target time** to be predicted and **Upstream water level, inflow to the reservoir, and weather data** from the previous 28 days
- **Output:** Predicted water level at the target time



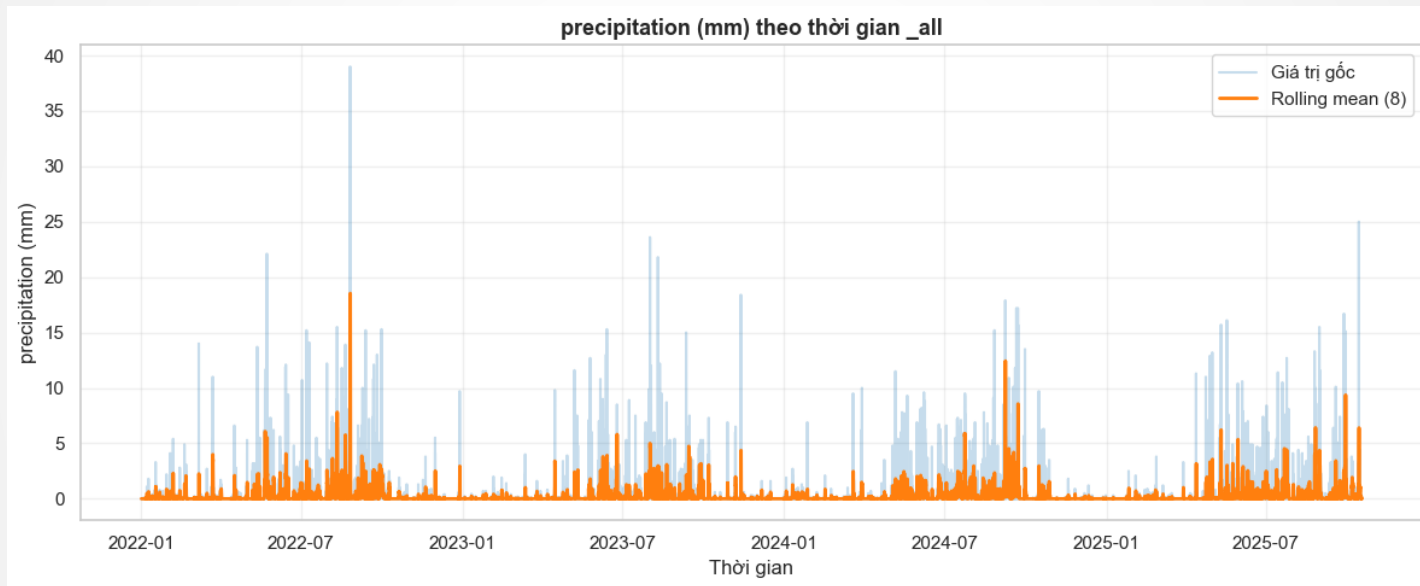
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EDA

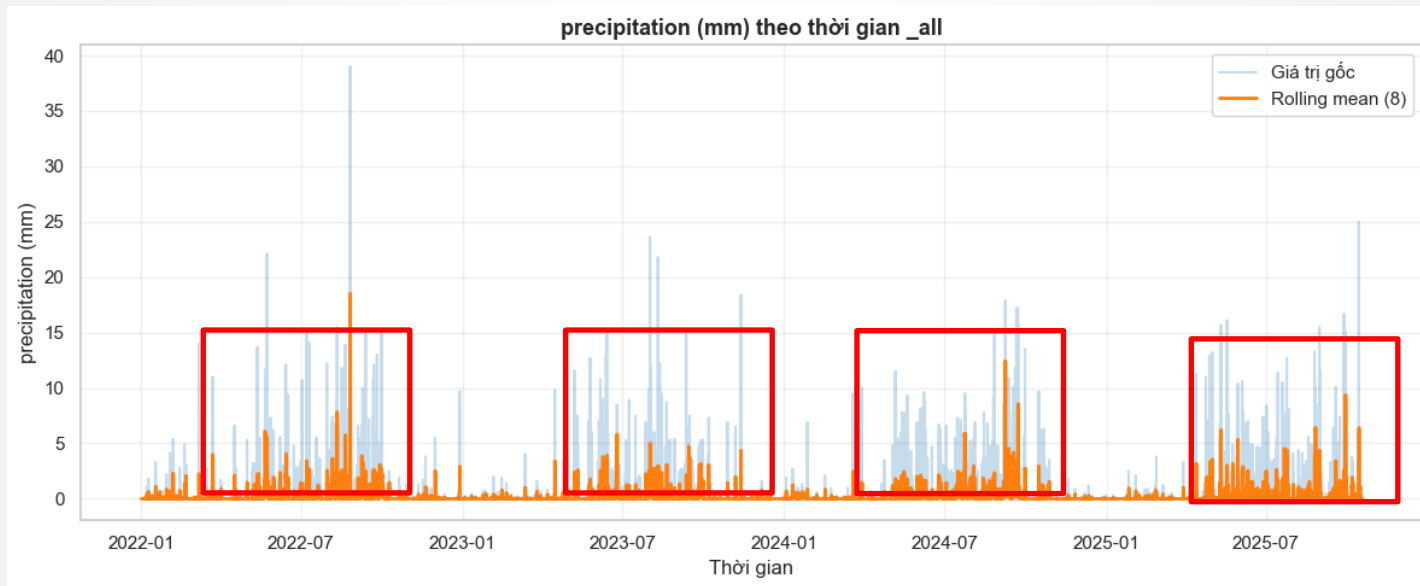
Contents of The weather data

Time	Date and time, representing the timestamp of each recorded observation.
precipitation_(mm)	Total amount of rainfall measured in millimeters during each hour.
temperature_2m_(°C)	Air temperature measured at 2 meters above the ground (°C).
relative_humidity_2m_(%)	Relative humidity (%) at 2 meters above the ground.
cloud_cover_(%)	Percentage of the sky covered by clouds (%).

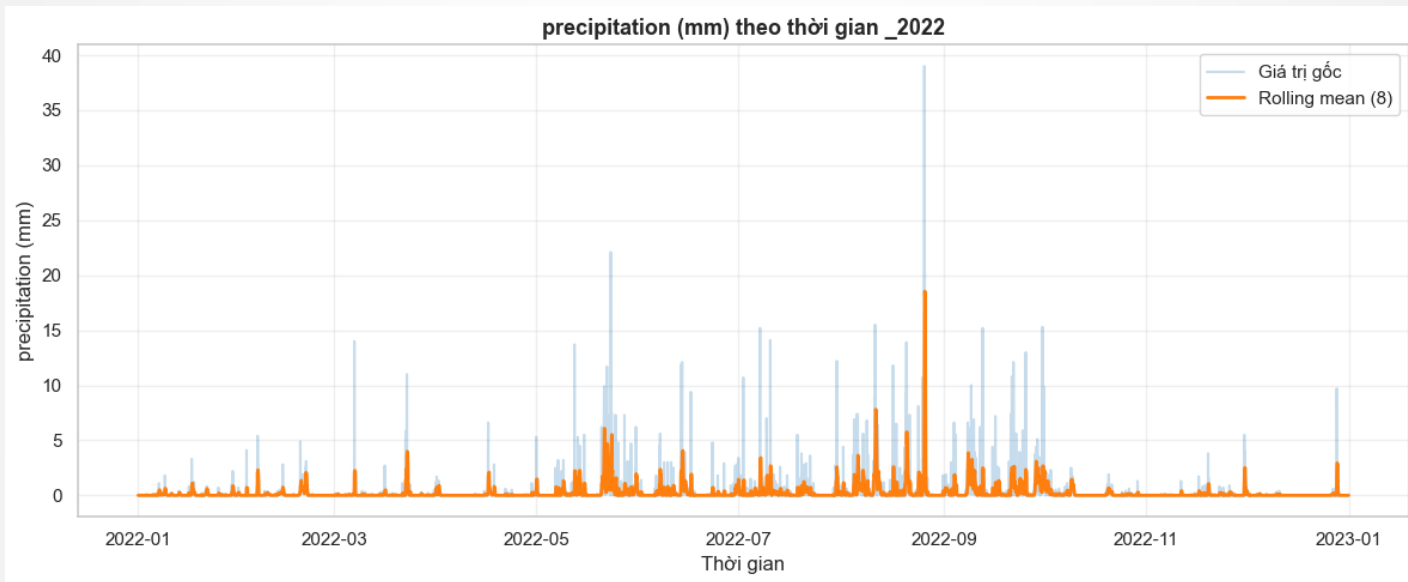
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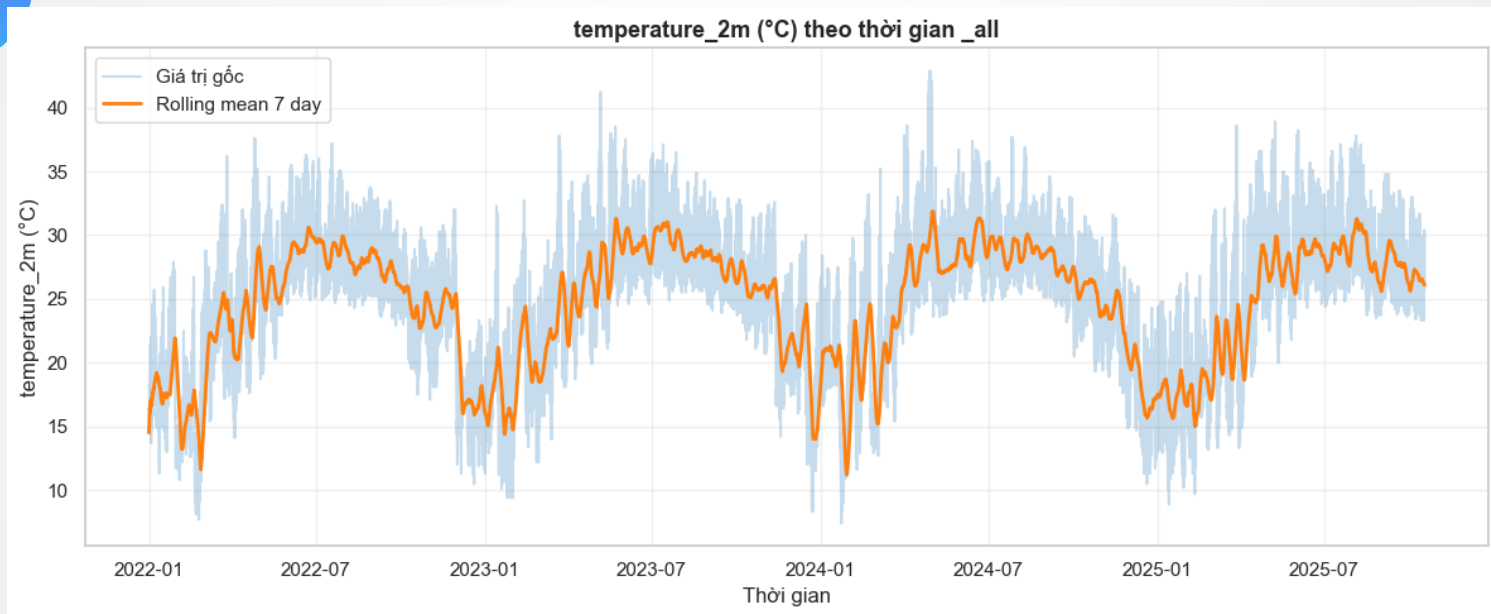
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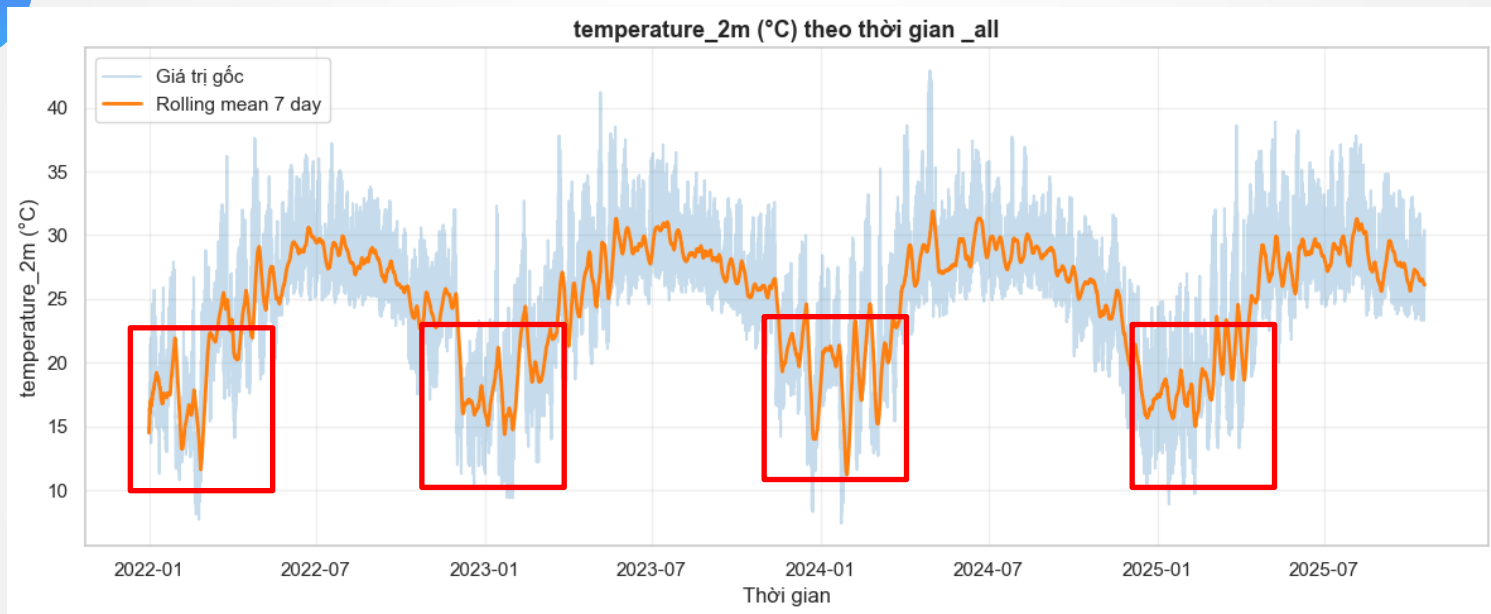
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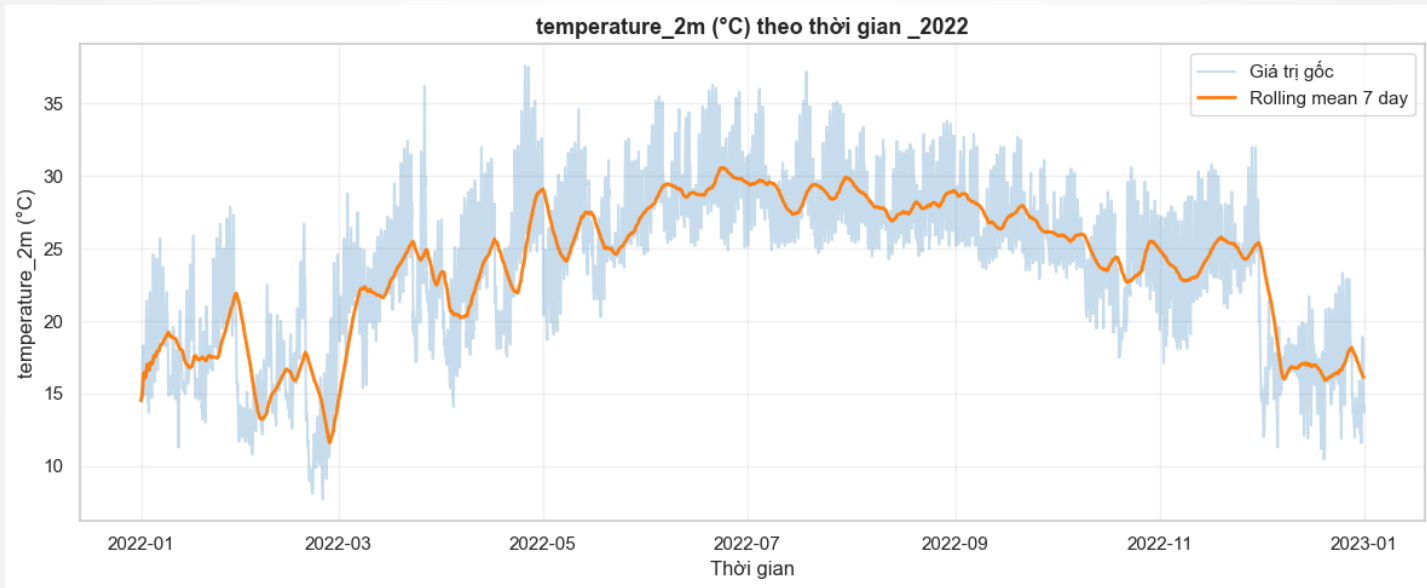
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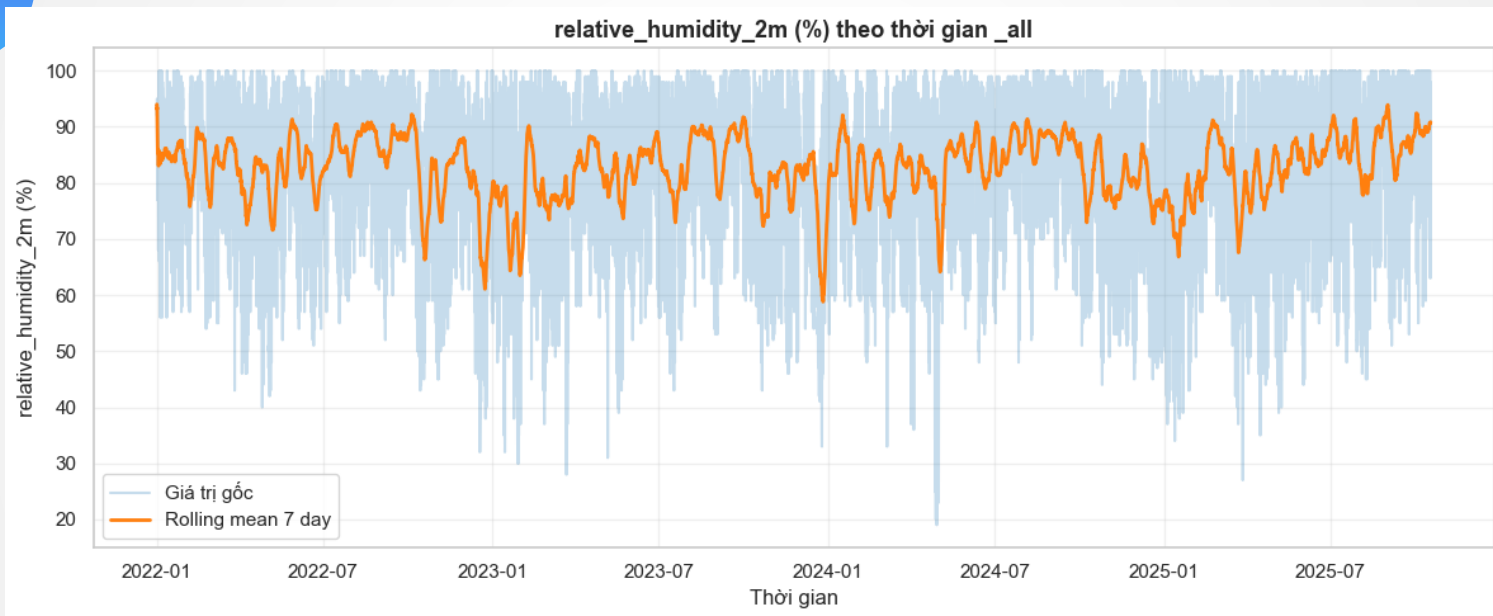
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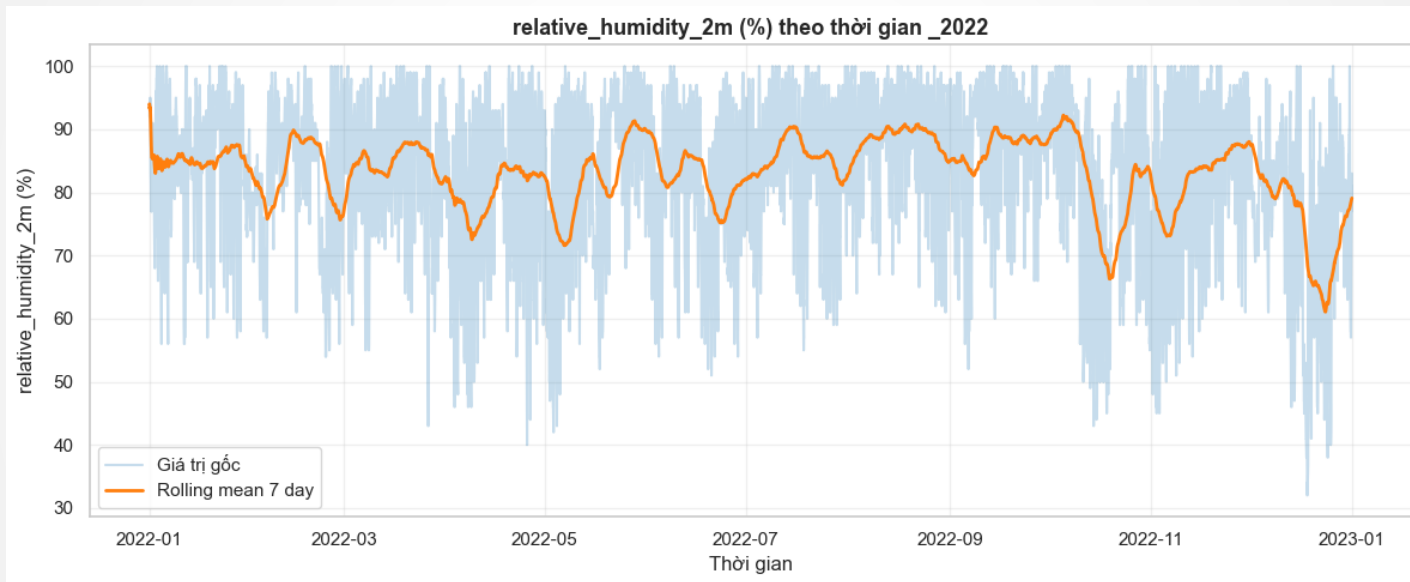
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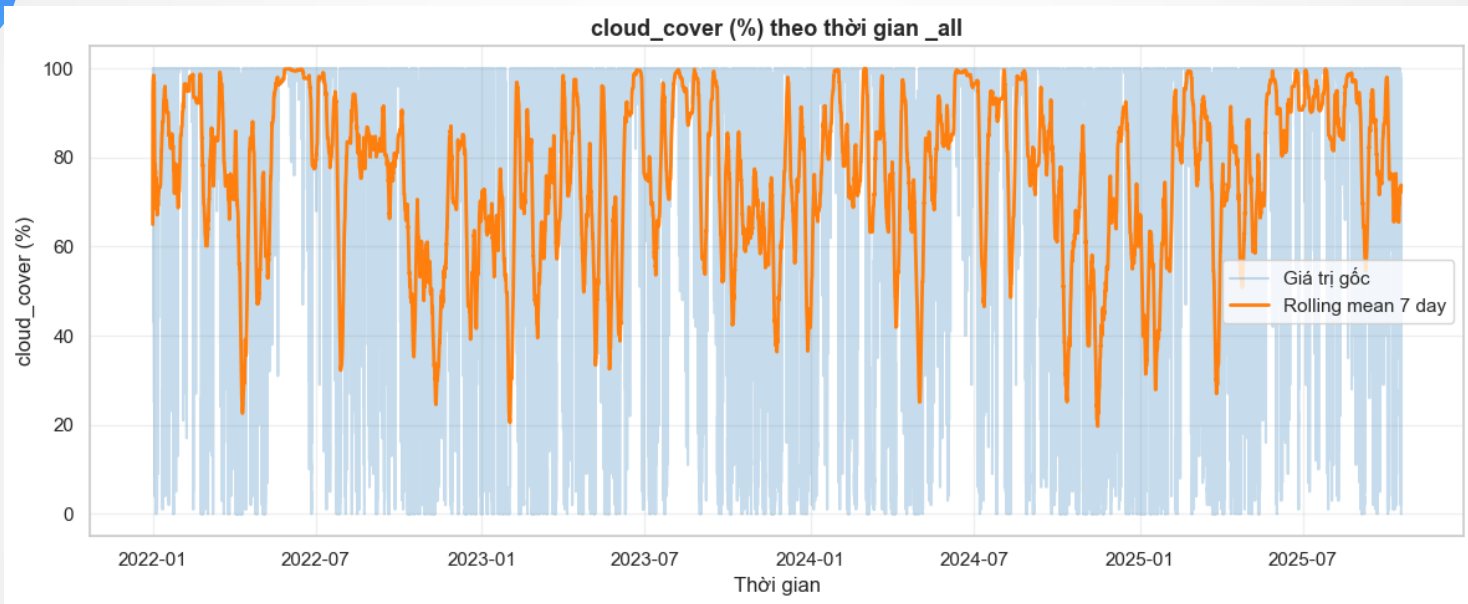
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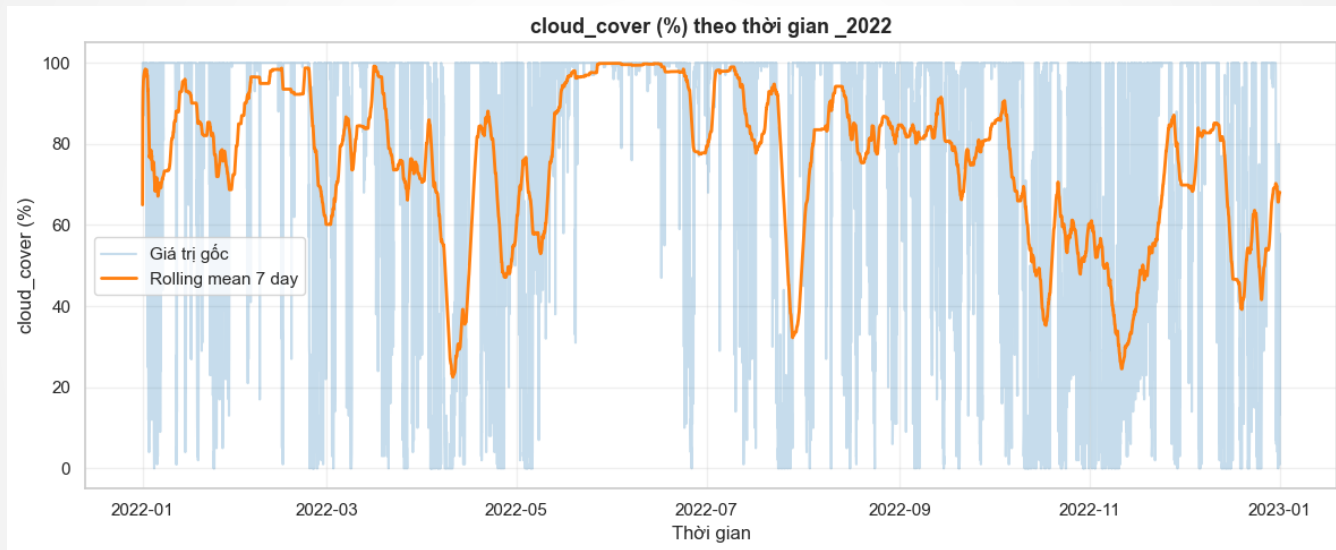
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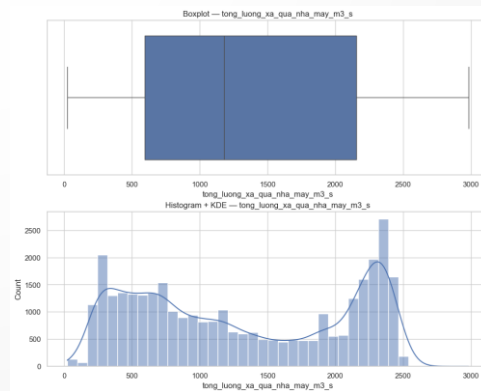
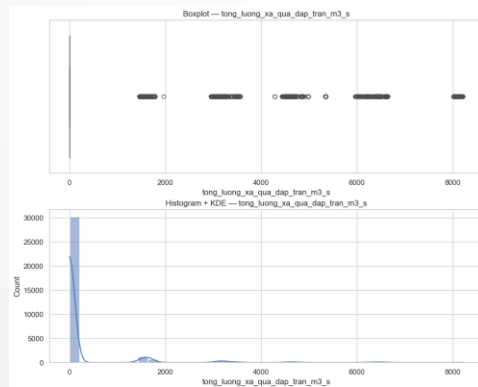
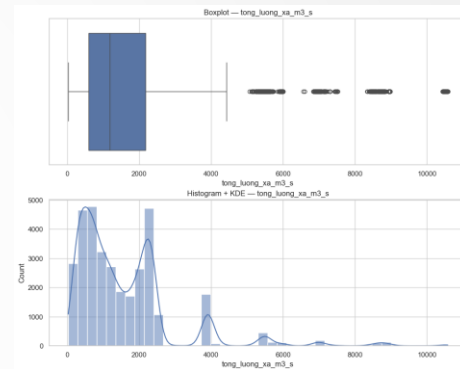
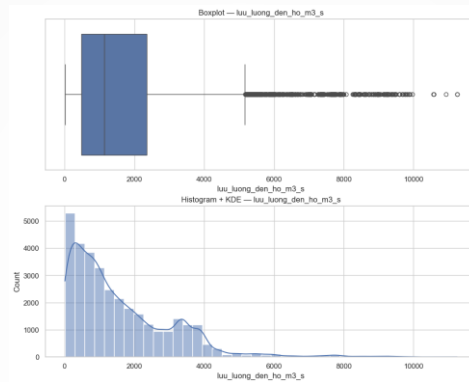
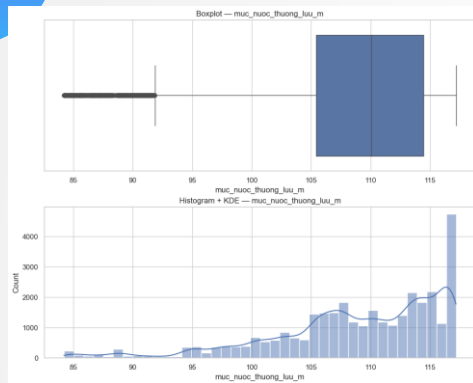
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muc_nuoc_chet_m	Lowest exploitable level of the reservoir
luu_luong_den_ho_m3_s	Volume of water entering the reservoir
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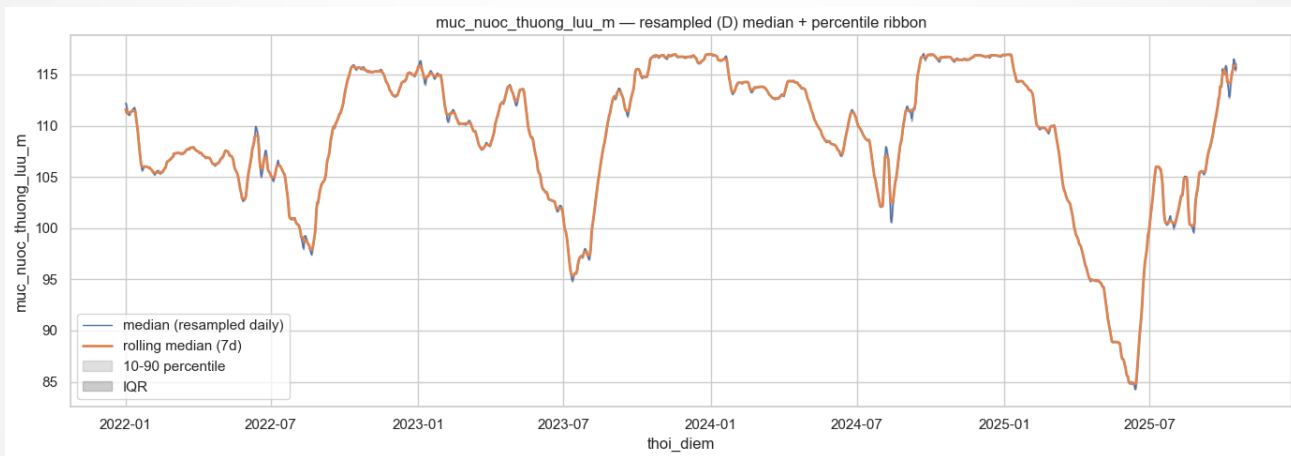
EDA

	Tên hồ	Thời điểm	Mức nước thượng lưu (m)	Mức nước dâng bình thường (m)	Mức nước chết (m)	Lưu lượng đến hồ (m3/s)	Tổng lượng xả (m3/s)	Tổng lượng xả qua đập tràn (m3/s)	Tổng lượng xả qua nhà máy (m3/s)	Số cửa xả sâu	Số cửa xả mặt
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1	Hòa Bình	1/1/2022 1:00	112.14	117	80	700	589	0	589	0	0
2	Hòa Bình	1/1/2022 2:00	112.14	117	80	700	589	0	589	0	0
3	Hòa Bình	1/1/2022 3:00	112.14	117	80	700	589	0	589	0	0
4	Hòa Bình	1/1/2022 4:00	112.14	117	80	700	589	0	589	0	0

EDA

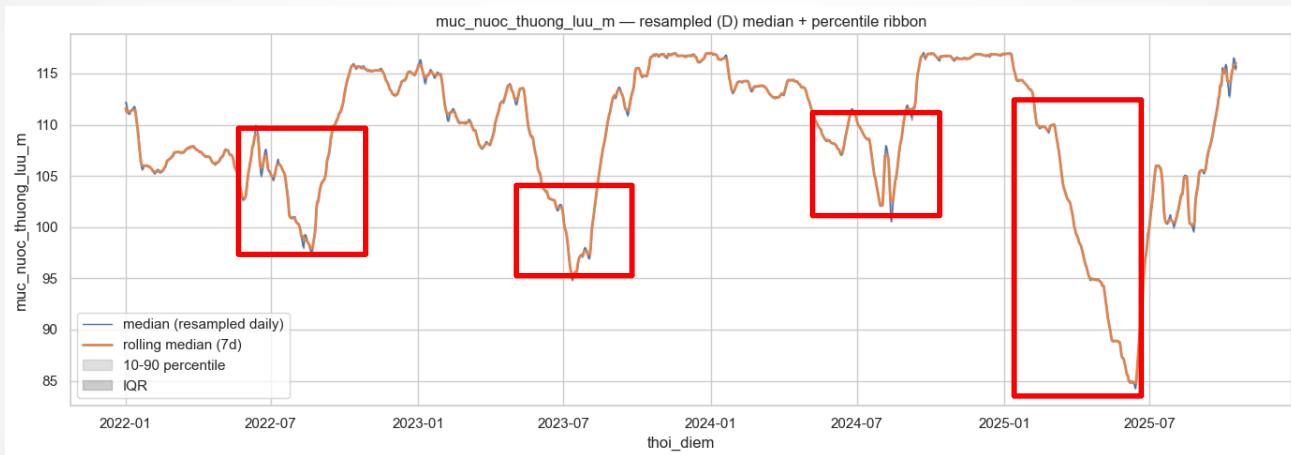


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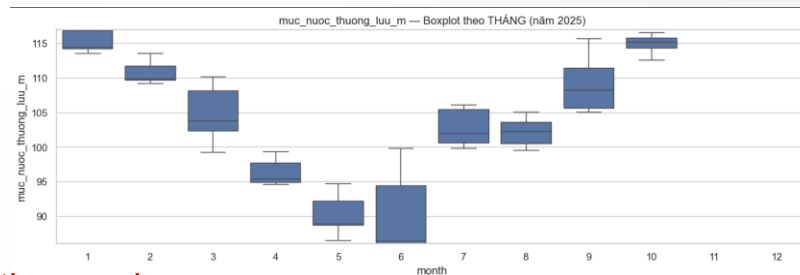
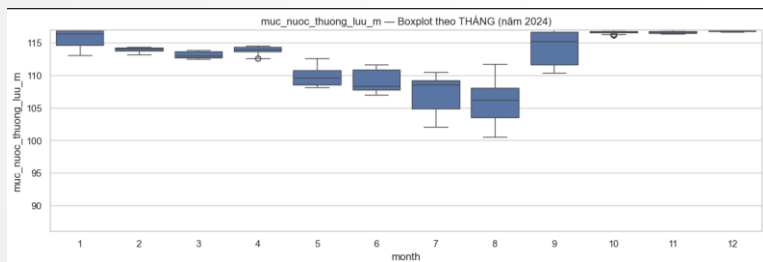
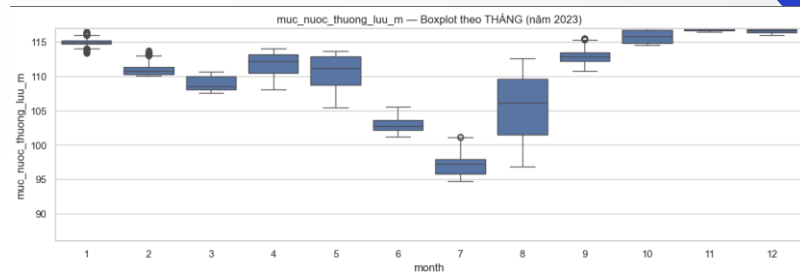
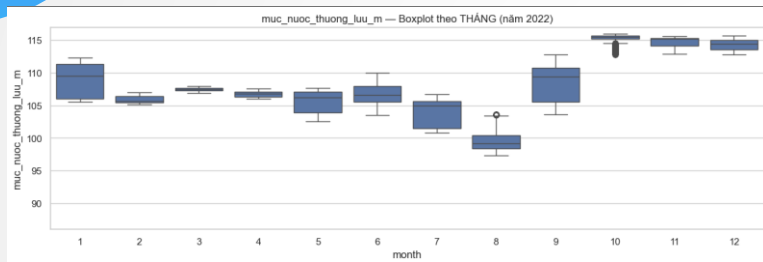
muc_nuoc_thuong_luu_m

EDA



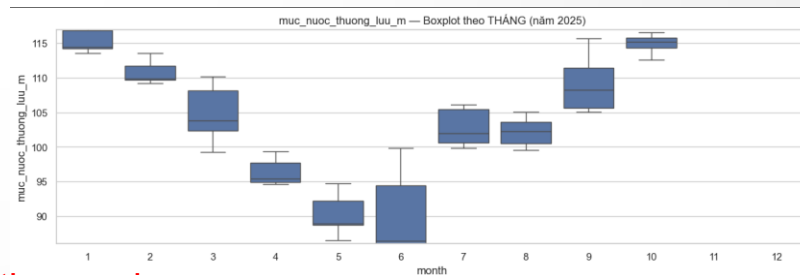
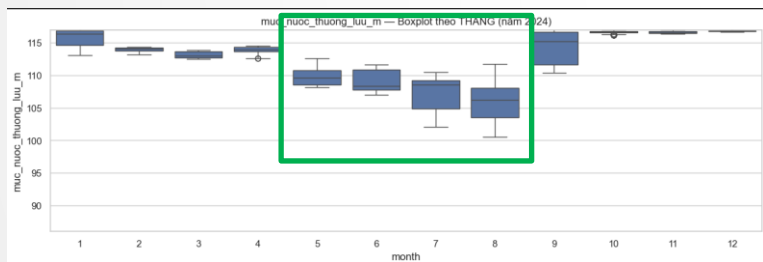
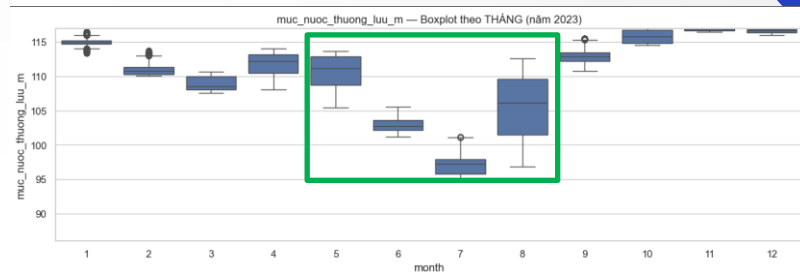
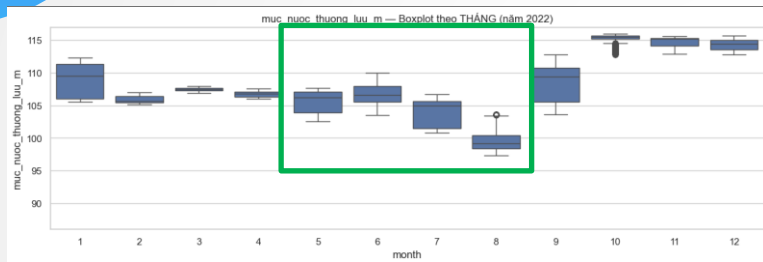
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EDA



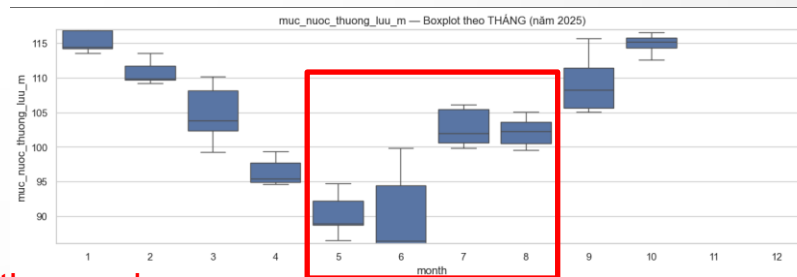
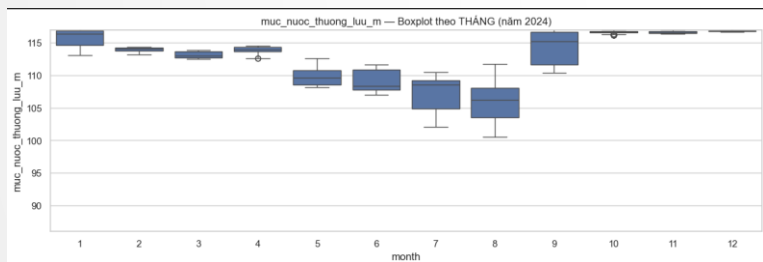
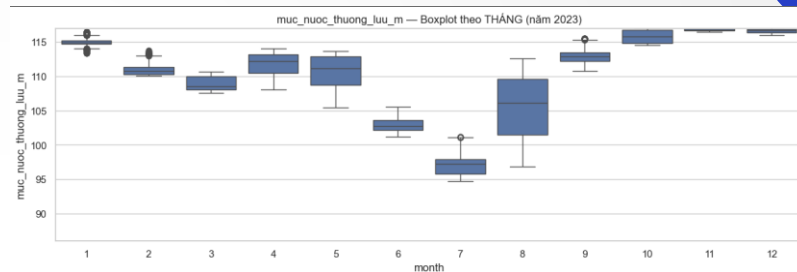
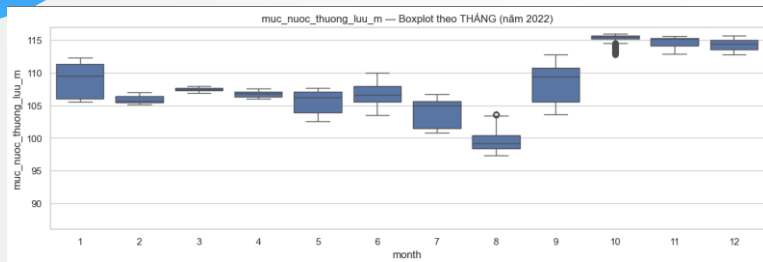
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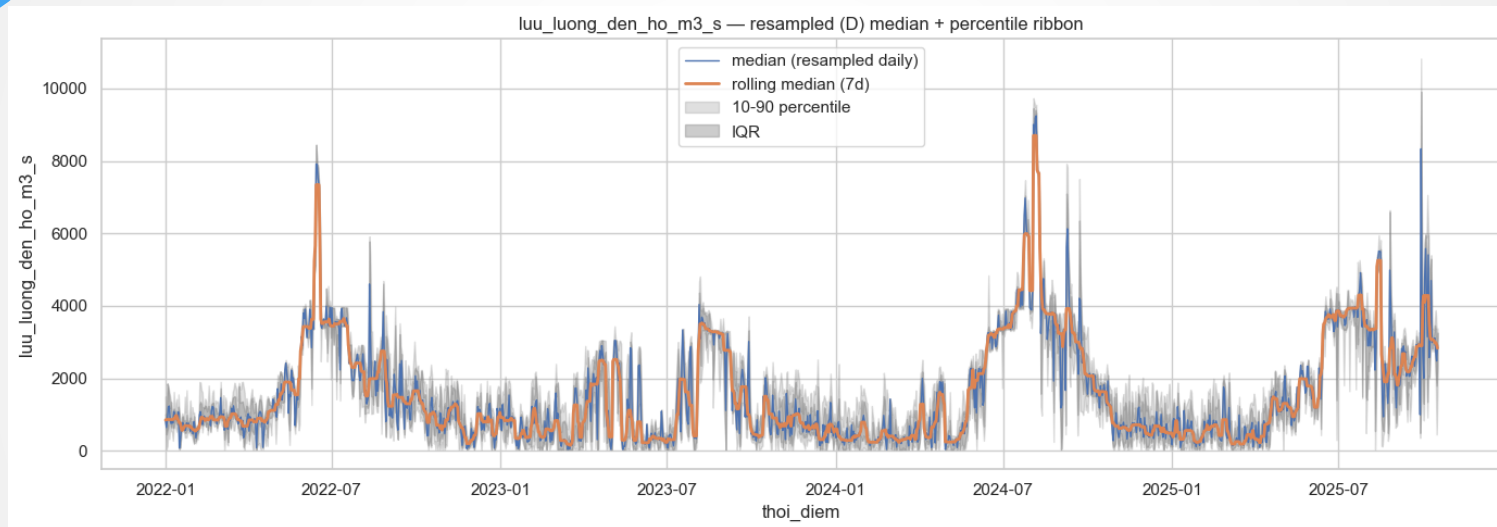
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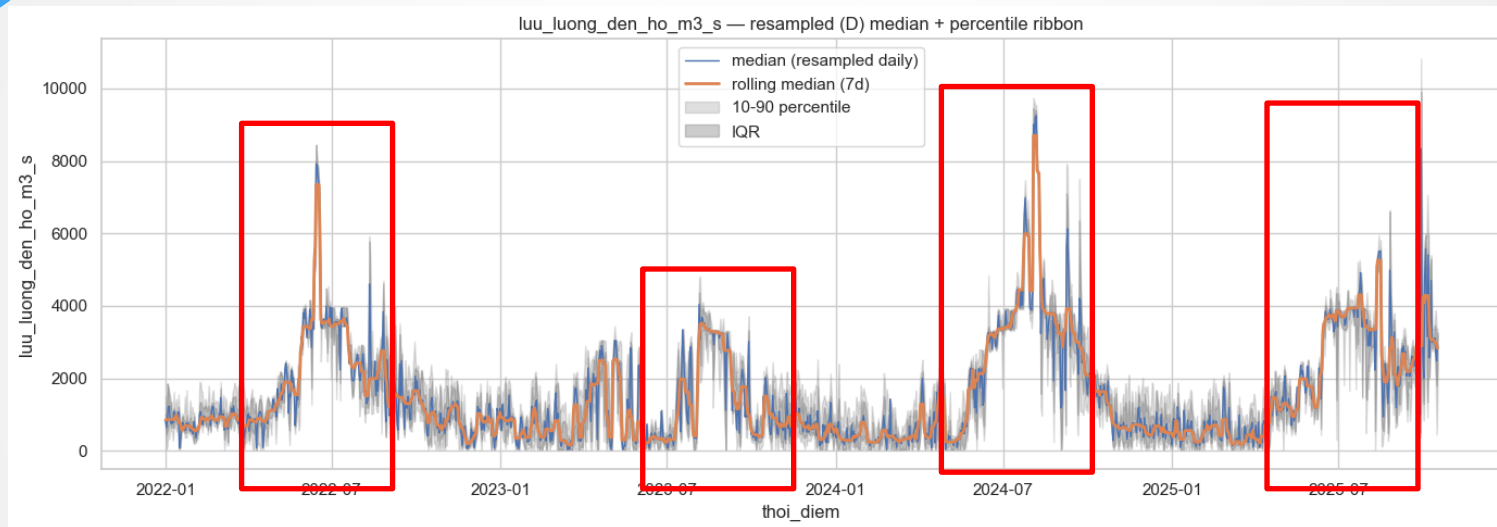
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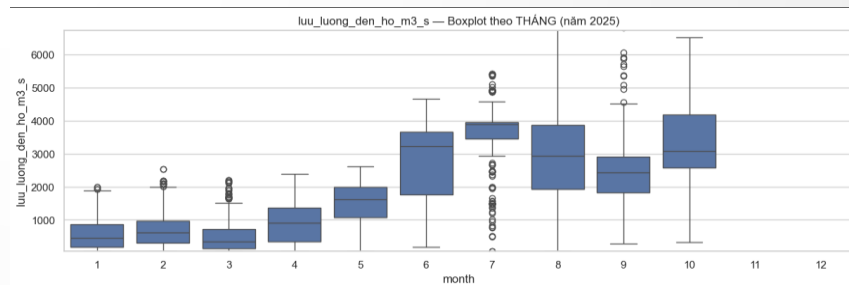
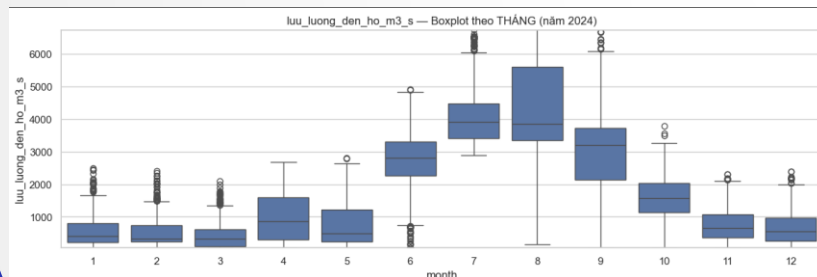
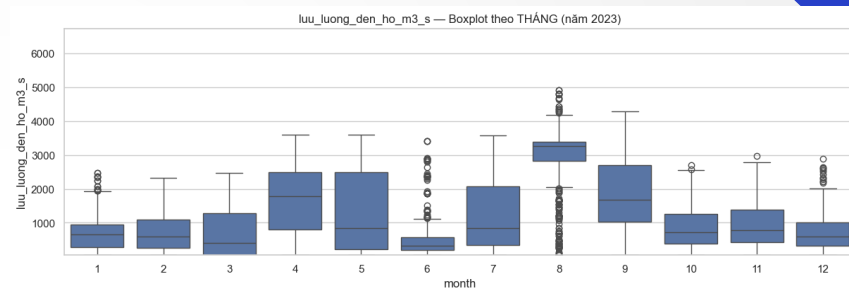
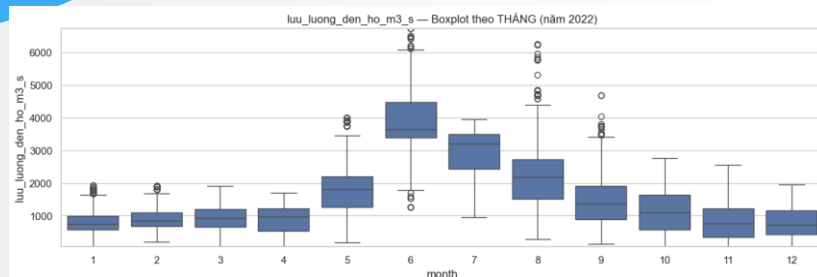
`luu_luong_den_ho_m3_s`

EDA



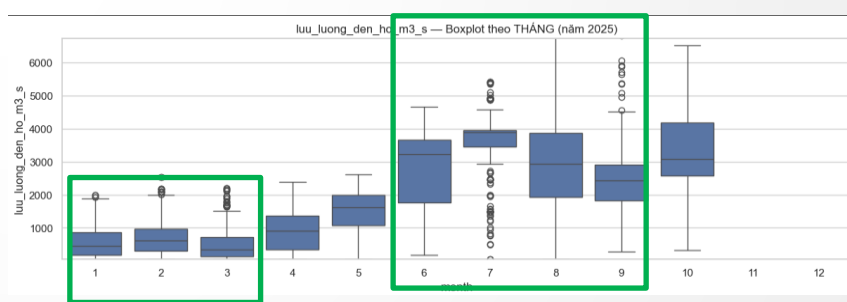
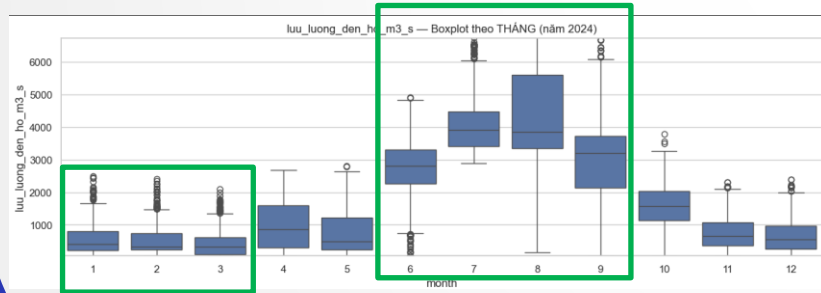
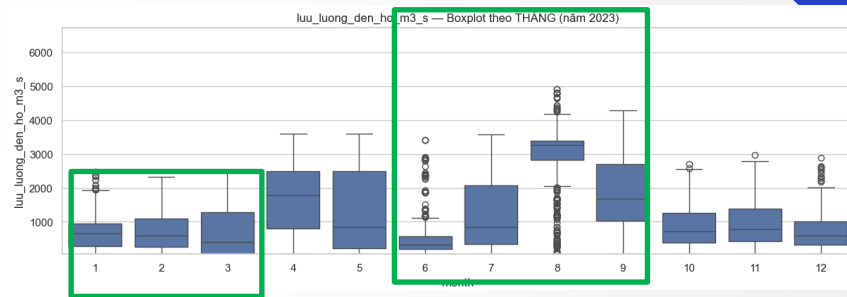
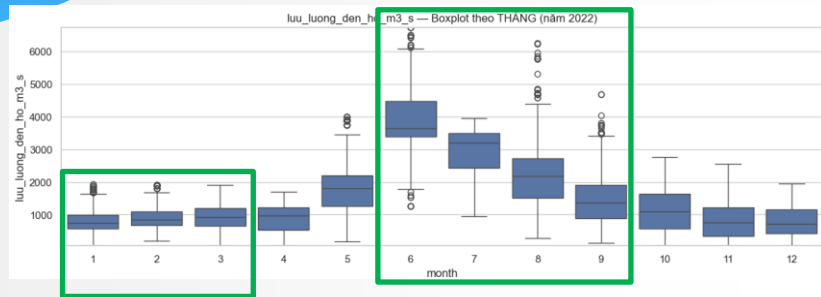
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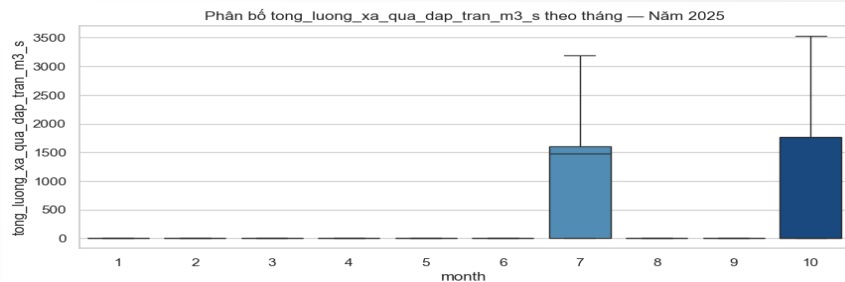
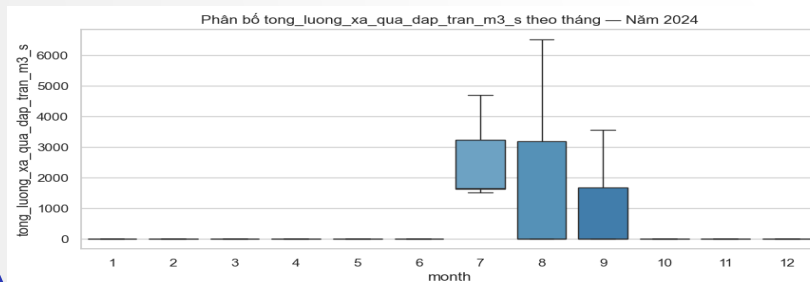
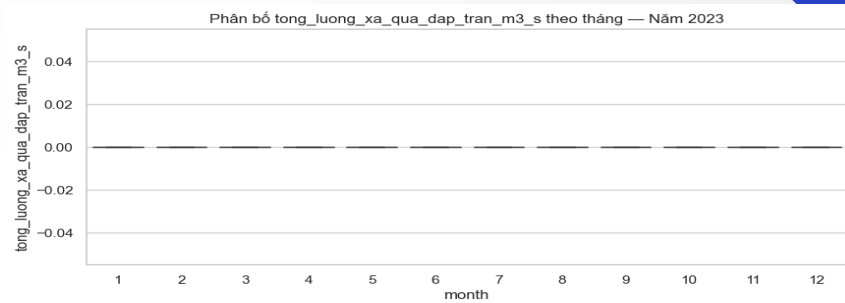
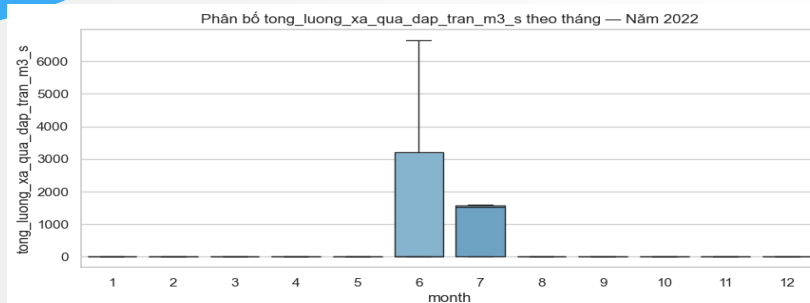
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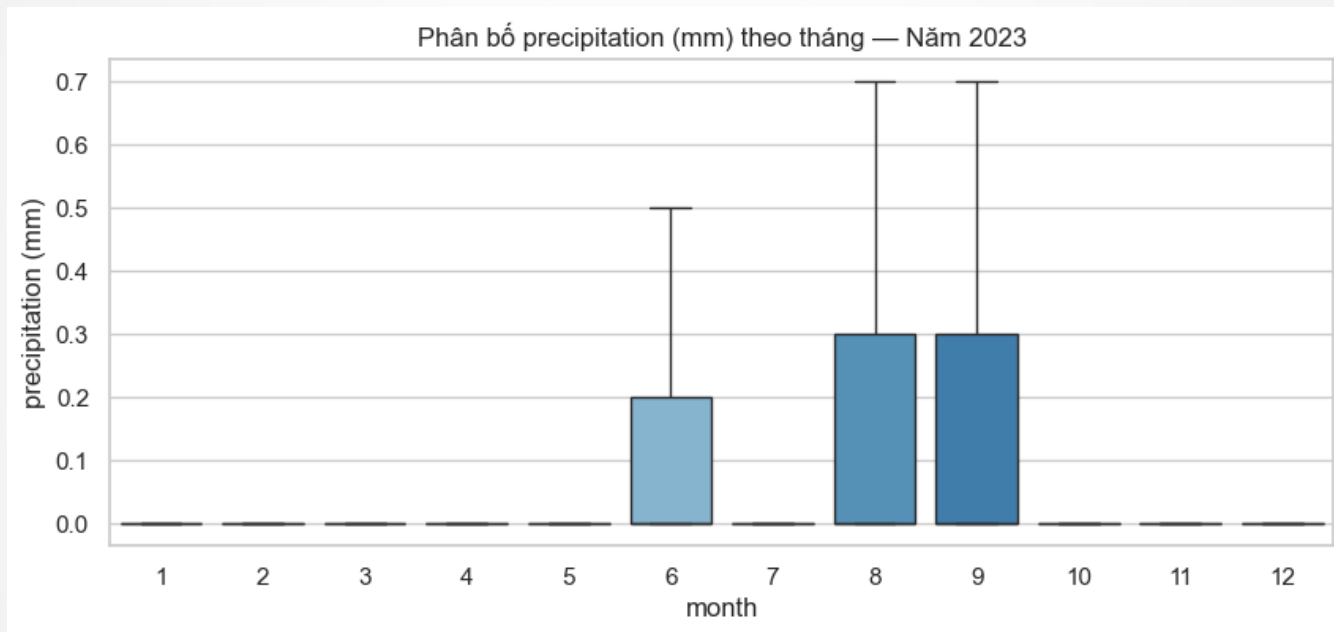
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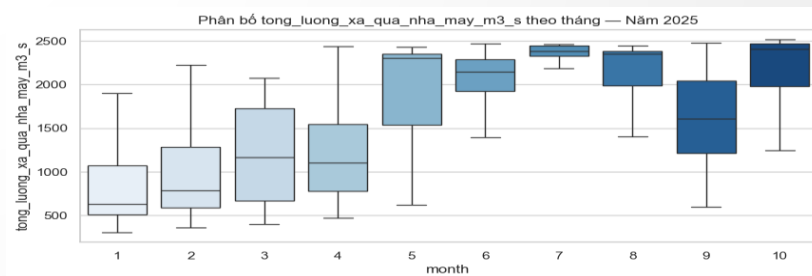
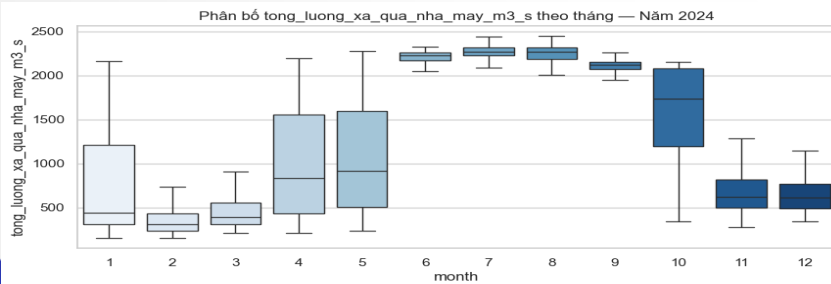
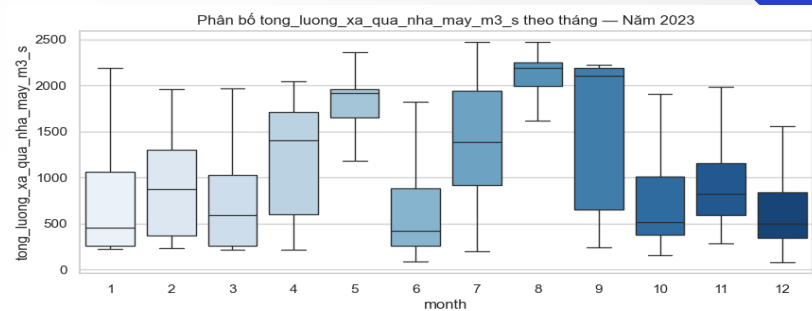
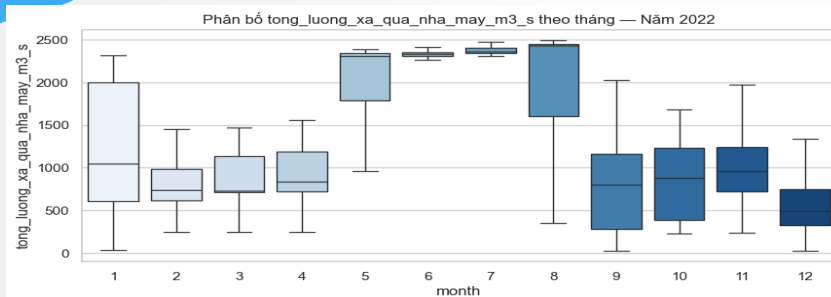


tong_luong_xa_qua_dap_tran_m3_s

EDA



EDA



tong_luong_xa_qua_nha_may_m3_s



03

Preprocessing

Contents of The Hoa Binh's Dam data

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luu_luong_den_ho_m3_s	Volume of water entering the reservoir
tong_luong_xa_m3_s	Total water released through all outlets
tong_luong_xa_qua_dap_tran_m3_s	Flood release; does not generate electricity
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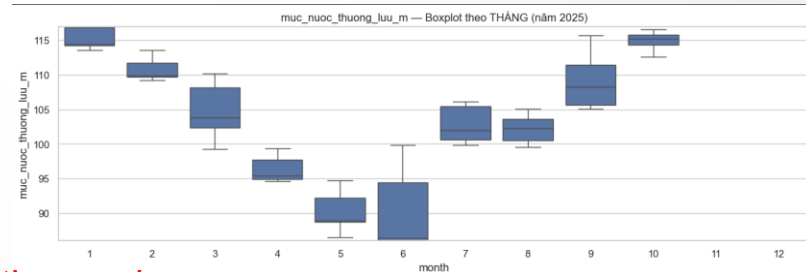
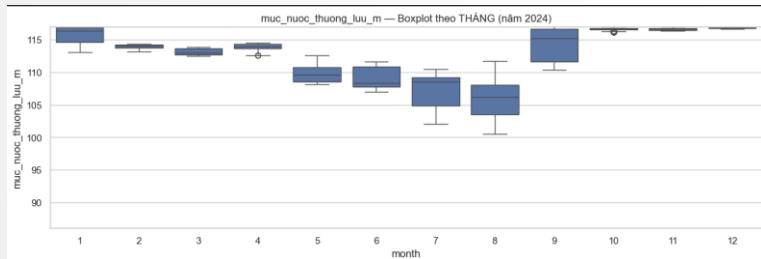
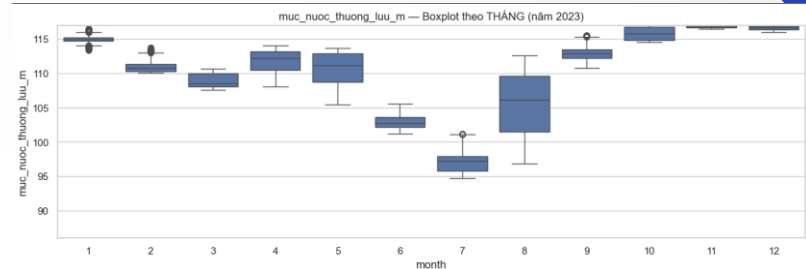
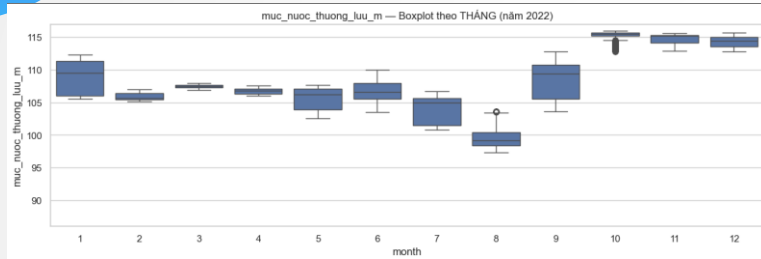
Preprocessing

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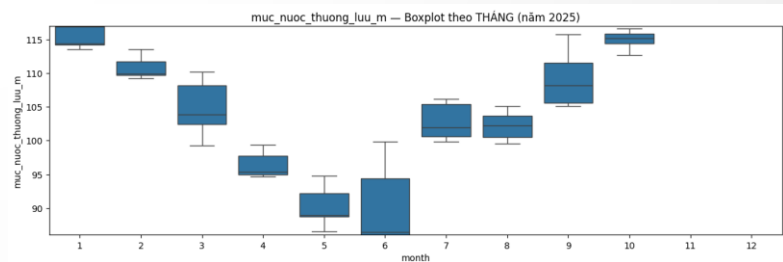
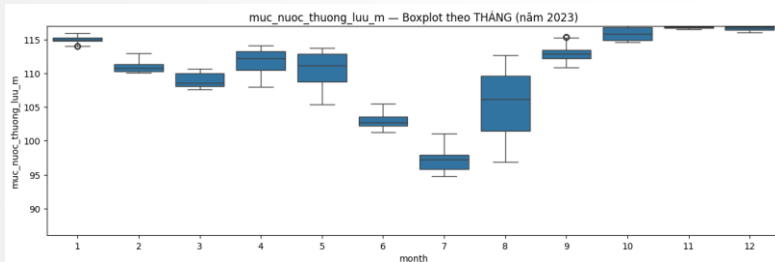
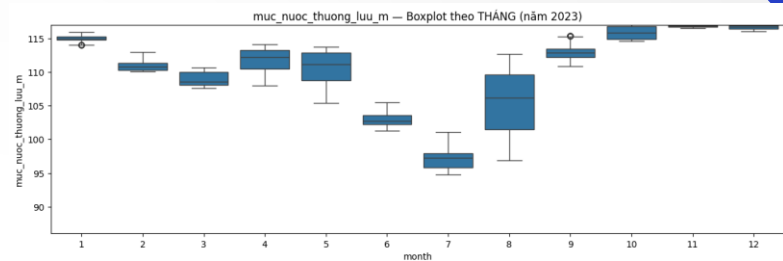
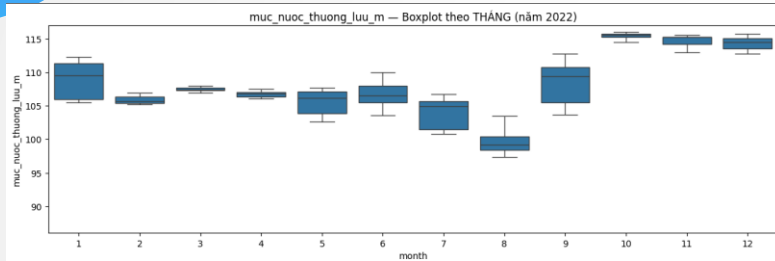
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Preprocessing



muc_nuoc_thuong_luu_m

Preprocessing



muc_nuoc_thuong_luu_m - preprocessed

Preprocessing

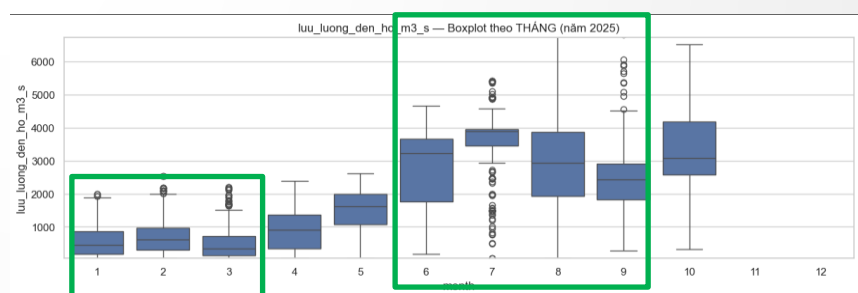
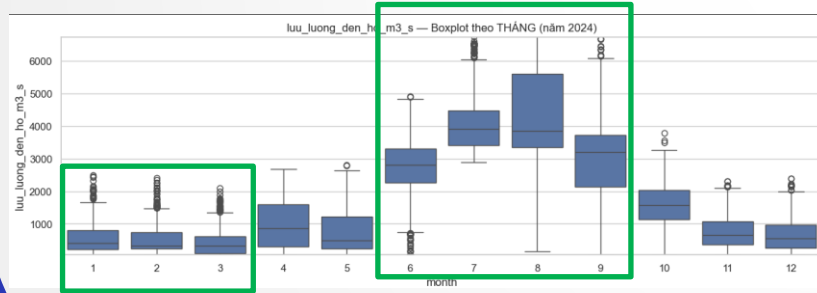
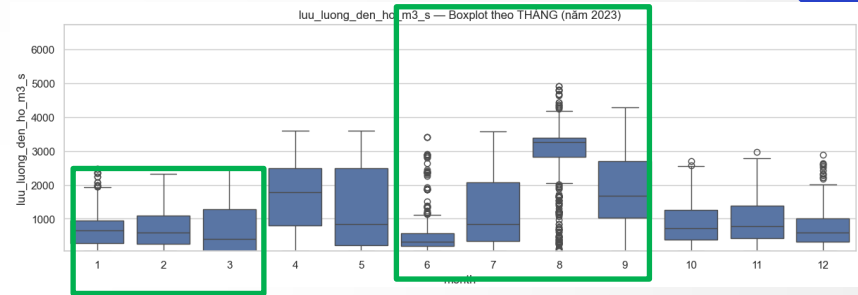
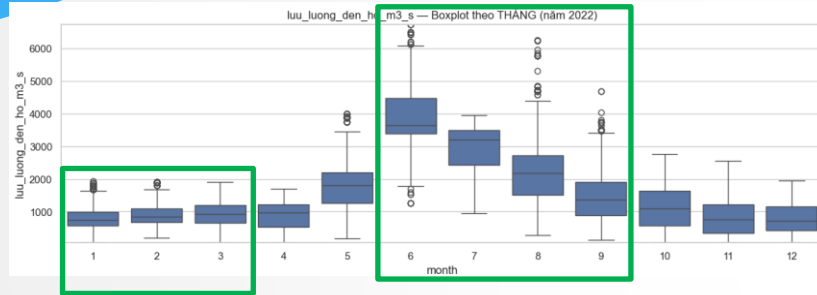
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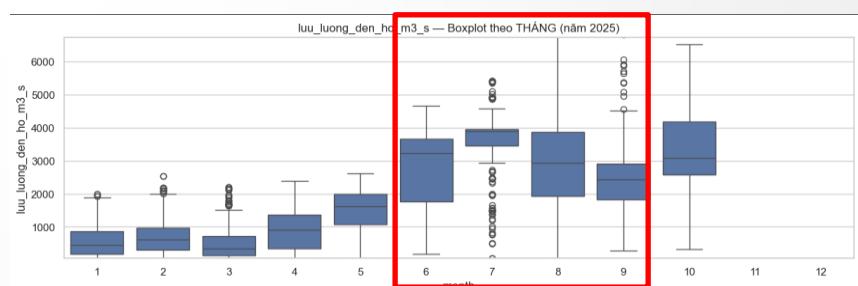
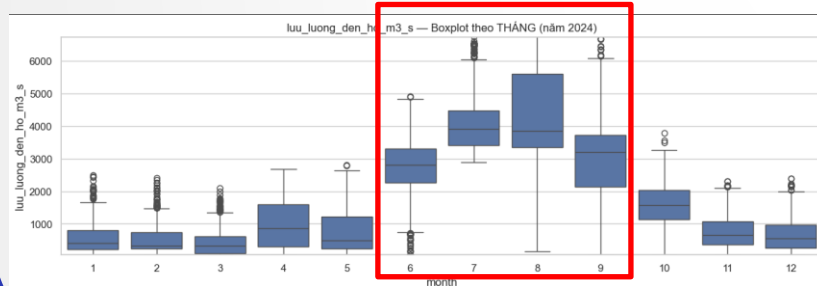
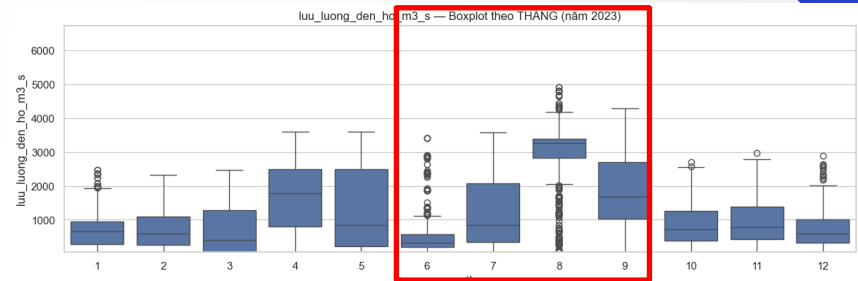
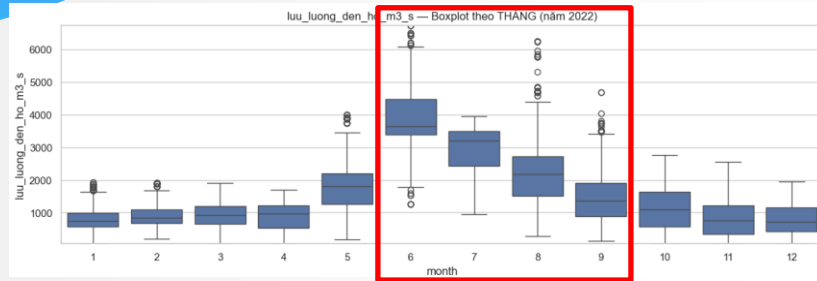
Processed in parallel during the data engineering step

Preprocessing



luu_luong_den_ho_m3_s

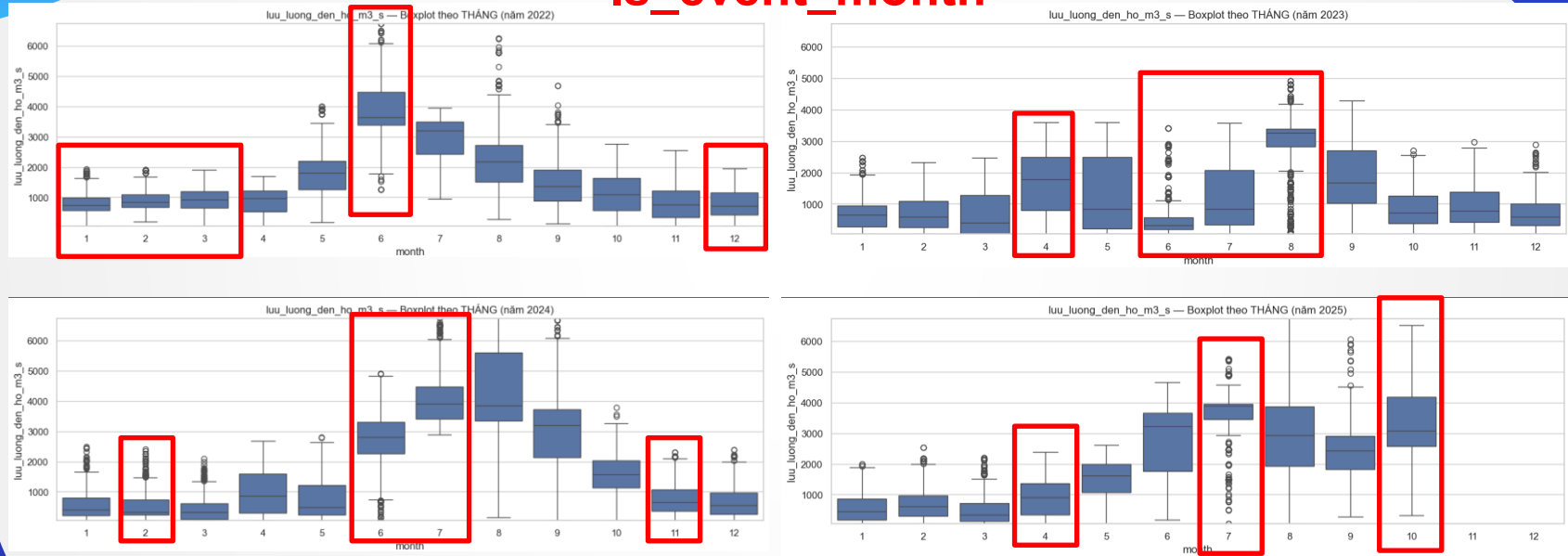
Preprocessing



luu_luong_den_ho_m3_s

Preprocessing

ls_event_month



luu_luong_den_ho_m3_s

Preprocessing

ls_event_month = 0

ls_event_month = 1

luu_luong_den_ho_m3_s

Preprocessing

Is_event_month = 0

Is_event_month = 1

- $(Q1, Q3, IQR = Q3 - Q1)$
- $\mathbf{LB} = \max(0, Q1 - 1.5 \times IQR)$
- $\mathbf{UB} = Q3 + 1.5 \times IQR$

Fill in the values outside $[LB, UB]$ with
LB/UB corresponding to **that month**.

luu_luong_den_ho_m3_s

Preprocessing

Is_event_month = 0

- $(Q1, Q3, IQR = Q3 - Q1)$
- $LB = \max(0, Q1 - 1.5 \times IQR)$
- $UB = Q3 + 1.5 \times IQR$

Fill in the values outside $[LB, UB]$ with LB/UB corresponding to **that month**.

Is_event_month = 1

Calculate on **log1p** to reflect multiplicative (right-tailed distribution):

- $x_l = \log(1 + x)$
- $Q1_l, Q3_l, IQR_l = Q3_l - Q1_l$
- $LB = \expm1(\max(0, Q1_l - 1.5 \times IQR_l))$
- $UB = \expm1(Q3_l + 2.5 \times IQR_l) \leftarrow$

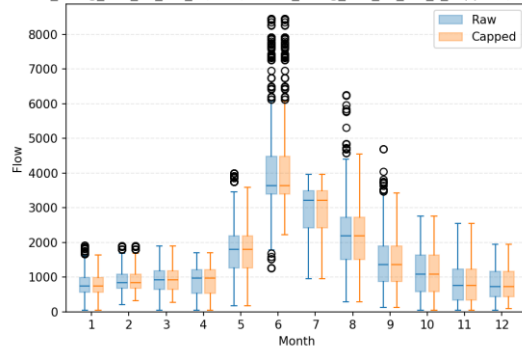
loosen the upper boundary to keep the real flood peak

Fill the values outside $[LB, UB]$ similarly.

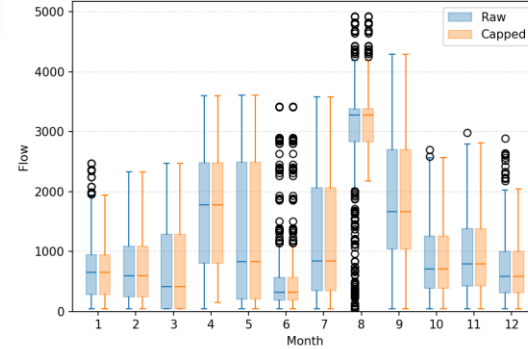
luu_luong_den_ho_m3_s

Preprocessing

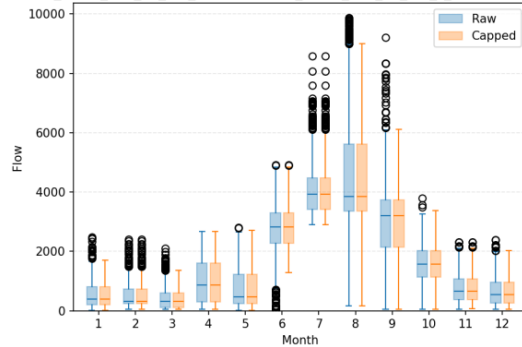
Monthly Boxplot (Raw vs Capped) — 2022
luu_luong_den_ho_m3_s (blue) vs luu_luong_den_ho_m3_s_capped (orange)



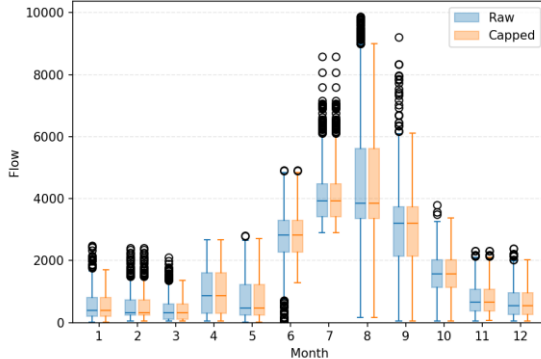
Monthly Boxplot (Raw vs Capped) — 2023
luu_luong_den_ho_m3_s (blue) vs luu_luong_den_ho_m3_s_capped (orange)



Monthly Boxplot (Raw vs Capped) — 2024
luu_luong_den_ho_m3_s (blue) vs luu_luong_den_ho_m3_s_capped (orange)



Monthly Boxplot (Raw vs Capped) — 2024
luu_luong_den_ho_m3_s (blue) vs luu_luong_den_ho_m3_s_capped (orange)



Preprocessing

thoi_diem	Timeline of hydroelectric reservoir operation data recording
muc_nuoc_thuong_luu_m	Indicates the current condition of the reservoir
luu_luong_den_ho_m3_s	Volume of water entering the reservoir
tong_luong_xa_qua_dap_tran_m3_s	Flood release; does not generate electricity
tong_luong_xa_qua_nha_may_m3_s	Used for electricity generation

Preprocessing

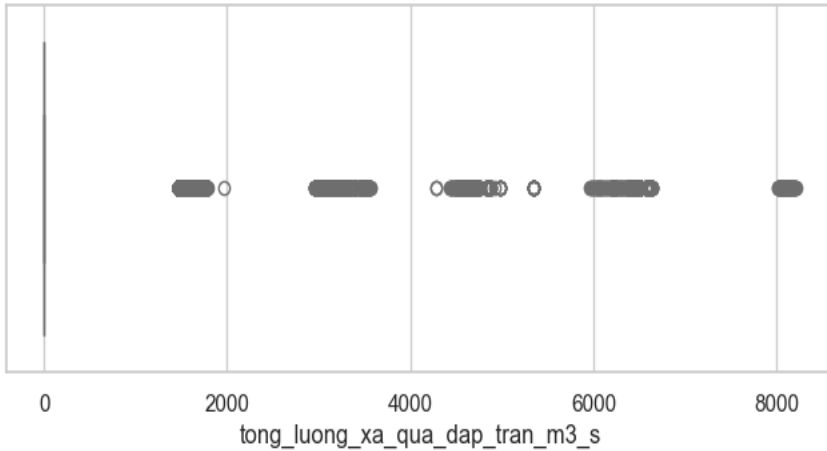
thoi_diem	Timeline of hydroelectric reservoir operation data recording
muc_nuoc_thuong_luu_m	Indicates the current condition of the reservoir
luu_luong_den_ho_m3_s	Volume of water entering the reservoir
tong_luong_xa_qua_dap_tran_m3_s	Flood release; does not generate electricity
tong_luong_xa_qua_nha_may_m3_s	Used for electricity generation

Preprocessing

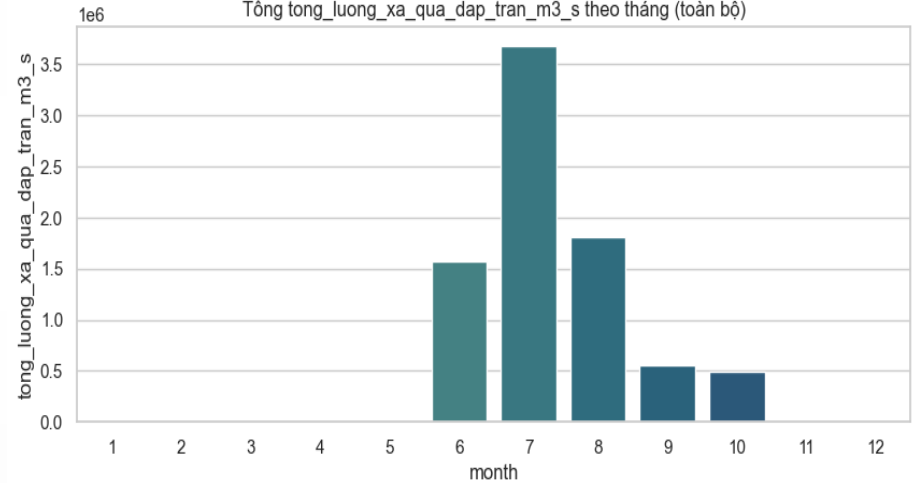
thoi_diem	Timeline of hydroelectric reservoir operation data recording
muc_nuoc_thuong_luu_m	Indicates the current condition of the reservoir
luu_luong_den_ho_m3_s	Volume of water entering the reservoir
tong_luong_xa_qua_dap_tran_m3_s	Flood release; does not generate electricity
tong_luong_xa_qua_nha_may_m3_s	Used for electricity generation

Preprocessing

Boxplot tong_luong_xa_qua_dap_tran_m3_s (Tổng thể)

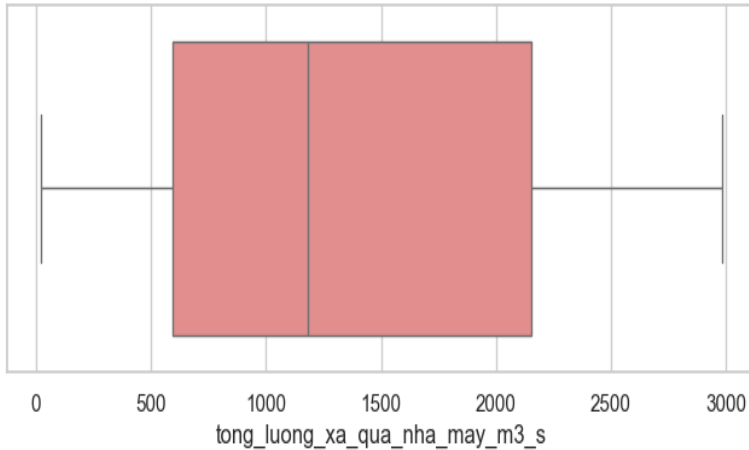


Tổng tong_luong_xa_qua_dap_tran_m3_s theo tháng (toàn bộ)

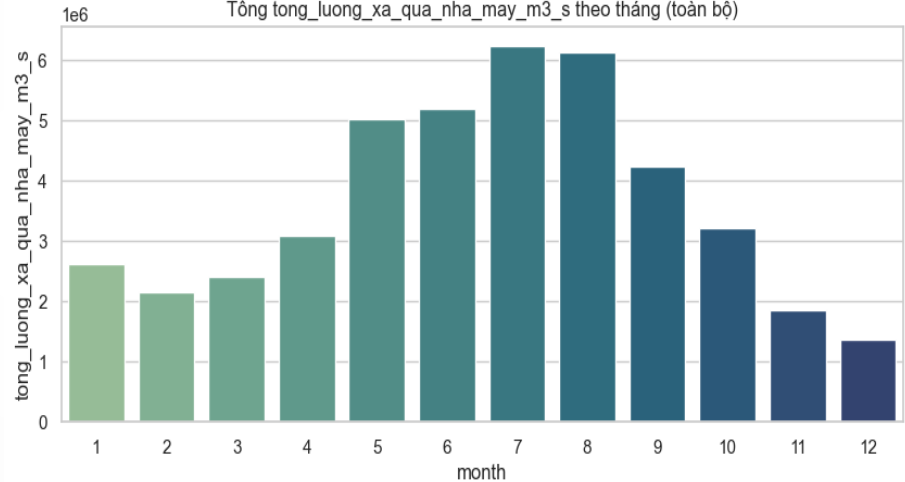


Preprocessing

Boxplot tong_luong_xa_qua_nha_may_m3_s (Tổng thể)



Tổng tong_luong_xa_qua_nha_may_m3_s theo tháng (toàn bộ)



Preprocessing

thoi_diem	Timeline of hydroelectric reservoir operation data recording
muc_nuoc_thuong_luu_m	Indicates the current condition of the reservoir
luu_luong_den_ho_m3_s	Volume of water entering the reservoir
dap_tran_norm	Normalized spillway discharge (0–1), calculated by dividing the actual discharge by the maximum value.
dap_tran_max	Maximum discharge capacity of the spillway (m ³ /s).
nha_may_norm	Normalized turbine discharge (0–1), calculated by dividing the actual discharge by the maximum value.
nha_may_max	Maximum discharge through the power plant turbines (m ³ /s).



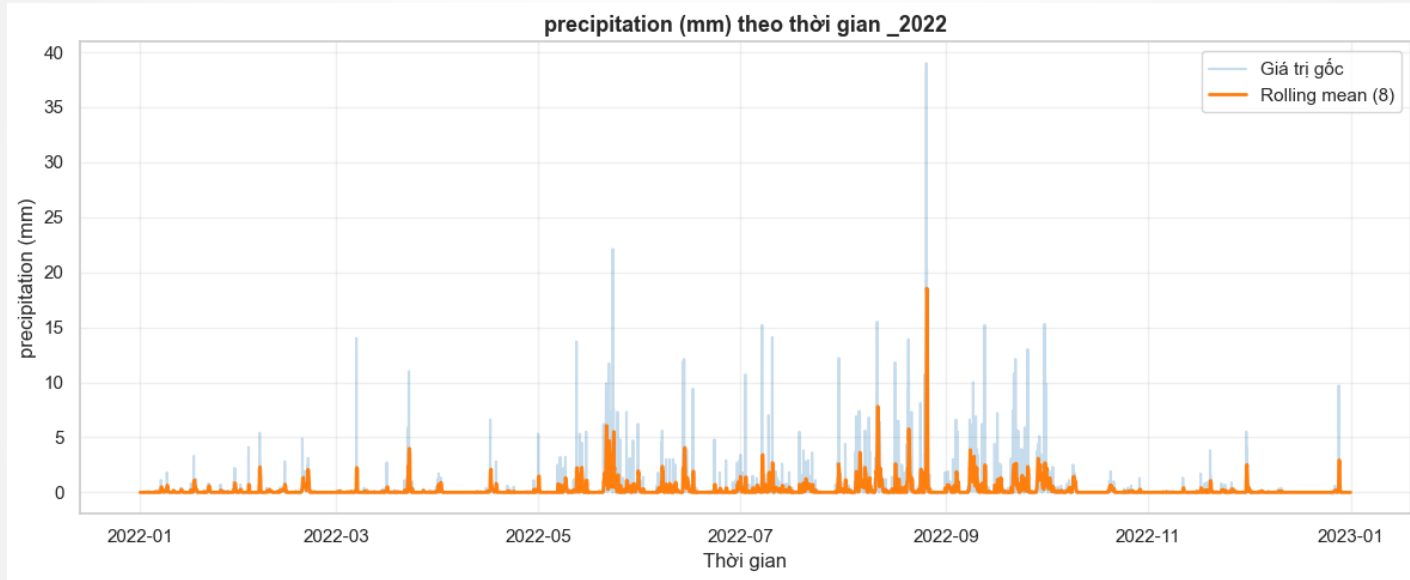
04

Feature engineering

Contents of The weather data

Time	Date and time, representing the timestamp of each recorded observation.
precipitation_(mm)	Total amount of rainfall measured in millimeters during each hour.
temperature_2m_(°C)	Air temperature measured at 2 meters above the ground (°C).
relative_humidity_2m_(%)	Relative humidity (%) at 2 meters above the ground.
cloud_cover_(%)	Percentage of the sky covered by clouds (%).

Feature engineering



Contents of The weather data

Time	Date and time, representing the timestamp of each recorded observation.
precipitation_(mm)	Total amount of rainfall measured in millimeters during each hour.
temperature_2m_(°C)	Air temperature measured at 2 meters above the ground (°C).
relative_humidity_2m_(%)	Relative humidity (%) at 2 meters above the ground.
cloud_cover_(%)	Percentage of the sky covered by clouds (%).
Season	Seasonal classification based on rainfall patterns: Rainy (May–October) and Dry (November–April).

Contents of The Hoa Binh's Dam data

thoi_diem	Timeline of hydroelectric reservoir operation data recording
muc_nuoc_thuong_luu_m	Indicates the current condition of the reservoir
luu_luong_den_ho_m3_s	Volume of water entering the reservoir
dap_tran_norm	Normalized spillway discharge (0–1), calculated by dividing the actual discharge by the maximum value.
dap_tran_max	Maximum discharge capacity of the spillway (m^3/s).
nha_may_norm	Normalized turbine discharge (0–1), calculated by dividing the actual discharge by the maximum value.
nha_may_max	Maximum discharge through the power plant turbines (m^3/s).

Contents of The Hoa Binh's Dam data

thoi_diem	Timeline of hydroelectric reservoir operation data recording
muc_nuoc_thuong_luu_m	Indicates the current condition of the reservoir
luu_luong_den_ho_m3_s	Volume of water entering the reservoir
dap_tran_norm	Normalized spillway discharge (0–1), calculated by dividing the actual discharge by the maximum value.
dap_tran_max	Maximum discharge capacity of the spillway (m ³ /s).
nha_may_norm	Normalized turbine discharge (0–1), calculated by dividing the actual discharge by the maximum value.
nha_may_max	Maximum discharge through the power plant turbines (m ³ /s).

Feature engineering

The Red River Basin Inter-Reservoir Process stipulates 3 flood season phases:

- **Early flood** (June 15–July 19): highest water level before flood ~ 105.0 m
- **Main flood** (July 20–August 21): reduced to 101.0 m
- **Late flood** (August 22–September 15): after September 16, water is allowed to gradually accumulate to 117 m → The reservoir is forced to lower from June–August to receive the flood → Flow distribution is significantly lower in these months.
- **Before June 26**: regulations on cutting off minor floods, the reservoir uses a capacity of up to 107 m to cut off early-season floods. → May–June may peak and then decrease rapidly.

Contents of The Hoa Binh's Dam data

thoi_diem	Timeline of hydroelectric reservoir operation data recording
muc_nuoc_thuong_luu_m	Indicates the current condition of the reservoir
luu_luong_den_ho_m3_s	Volume of water entering the reservoir
tong_luong_xa_m3_s	Total water released through all outlets
tong_luong_xa_qua_dap_tran_m3_s	Flood release; does not generate electricity
tong_luong_xa_qua_nha_may_m3_s	Used for electricity generation
phase	Flood season stages

Contents of The Hoa Binh's Dam data

<code>thoi_diem</code>	Timeline of hydroelectric reservoir operation data recording
<code>muc_nuoc_thuong_luu_m</code>	Indicates the current condition of the reservoir
<code>luu_luong_den_ho_m3_s</code>	Volume of water entering the reservoir
<code>dap_tran_norm</code>	Normalized spillway discharge (0–1), calculated by dividing the actual discharge by the maximum value.
<code>dap_tran_max</code>	Maximum discharge capacity of the spillway (m ³ /s).
<code>nha_may_norm</code>	Normalized turbine discharge (0–1), calculated by dividing the actual discharge by the maximum value.
<code>nha_may_max</code>	Maximum discharge through the power plant turbines (m ³ /s).

Feature engineering

Detecting “Event month”:

- With the aim of identifying months with extreme or unusual volatility, to **loosen the capping margin** (not cutting off valid peaks).

Feature engineering

Detecting “Event month”:

- **$\Delta\%$ month-over-month** in each year: large $\rightarrow$ strong volatility ($\geq 150\%$).
- **Climate deviation (≥ 1.5):**
$$\text{anom_iqr_units} = \frac{\text{median}_{y,m} - \text{median}_{\text{climatology_month}}}{\text{IQR}_{\text{climatology_month}}}$$
- **Outlier % in the month** (according to IQR $1.5 \times$) ($\geq 8\%$).

Contents of The Hoa Binh's Dam data

thoi_diem	Timeline of hydroelectric reservoir operation data recording
muc_nuoc_thuong_luu_m	Indicates the current condition of the reservoir
luu_luong_den_ho_m3_s	Volume of water entering the reservoir
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nha_may_max	Maximum discharge through the power plant turbines (m ³ /s).
phase	Flood season stages
luu_luong_den_ho_m3_s_capped_month_median	Monthly crop platform
...	

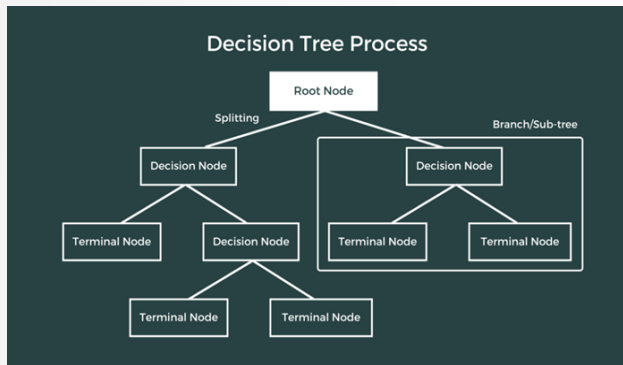
Contents of The Hoa Binh's Dam data

...	
luu_luong_den_ho_m3_s_capped_pct_in_month	Relative position in the month
pct_change	Sudden fluctuations between months ($\geq 150\%$)
anom_iqr_units	Climate deviation (median/IQR) (≥ 1.5)
out_pct	Extreme value ratio in the month ($\geq 8\%$)
is_event_month	Event Background

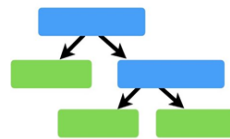
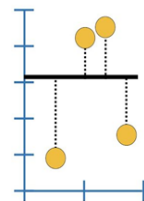
05

Modeling

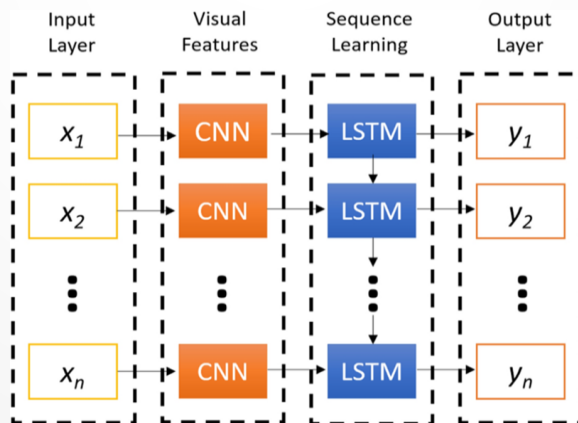
Modeling



XGBoost Part 1:



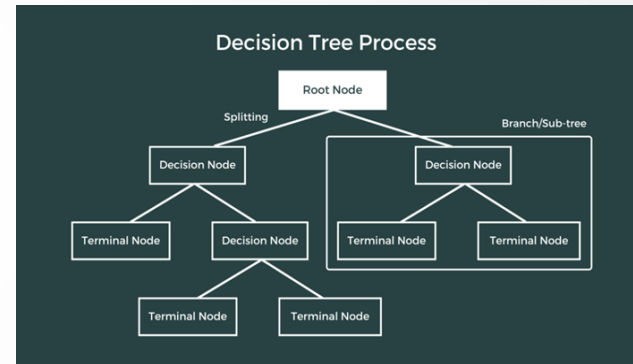
XGBoost Trees for Regression!!!



Modeling

Decision Tree — Interpretable Baseline:

- Provides a **simple** and **explainable benchmark** for nonlinear or threshold-based relationships (e.g., water level, inflow, rainfall) with very low training cost, allowing quick validation of whether the data and features are informative.

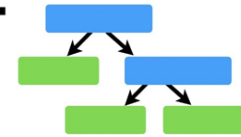
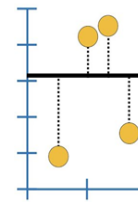


Modeling

XGBoost — Strong Standard for Tabular Data:

- Delivers high and stable performance on tabular datasets with many engineered features over a 28-day window (lags, rolling statistics, quantiles). It effectively captures nonlinearities and interactions between hydrological and weather variables, while its built-in regularization helps prevent overfitting.

XGBoost Part 1:

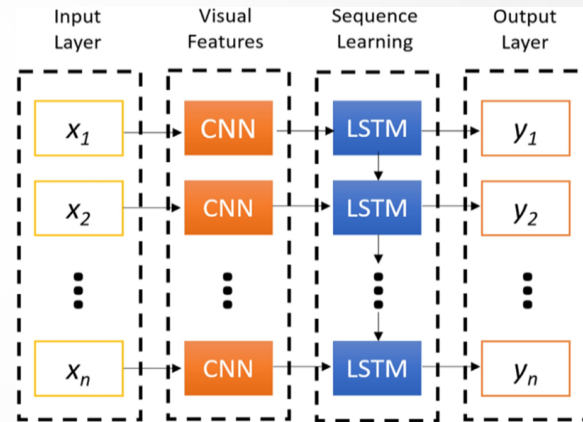


XGBoost Trees for Regression!!!

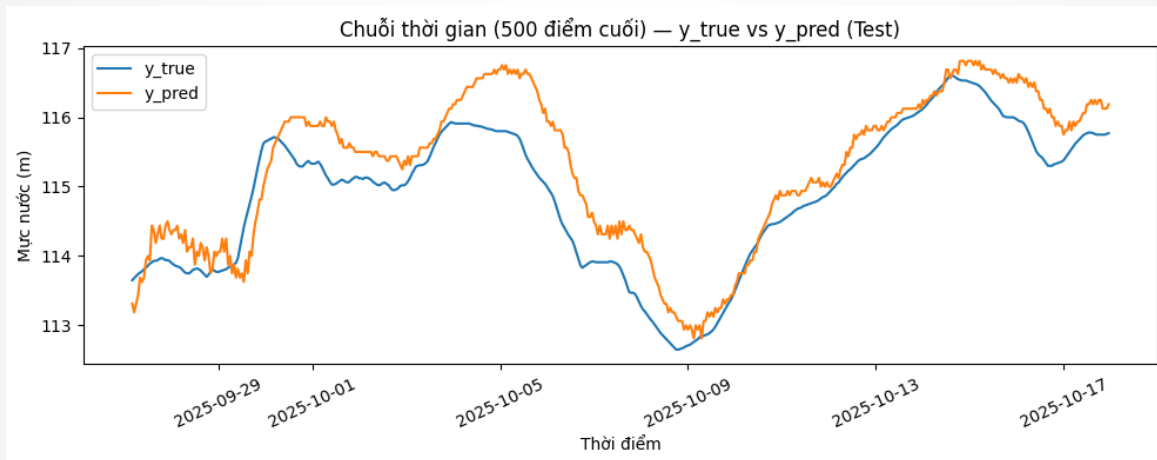
Modeling

CNN+LSTM — Specialist for Multivariate Time Series:

- Directly models the temporal patterns and dependencies within the 28-day sequence — CNN extracts local patterns (e.g., rainfall spikes, flow surges), while LSTM captures long-term and delayed dependencies — enabling the model to learn dynamic behaviors that tree-based models cannot capture without handcrafted features.

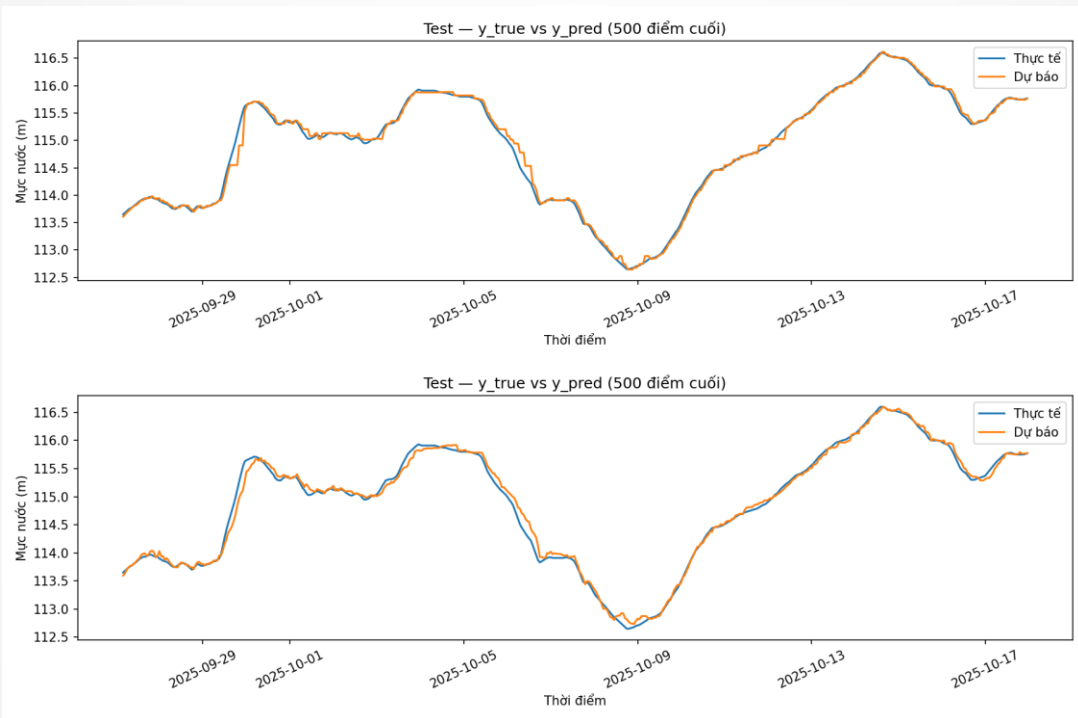


Modeling



CNN + LSTM

Modeling



Descision Tree

XGBoost

Metric Summary

Method	N	MAE	RMSE	R2
Decision Tree	6,653	1.3386	3.3292	0.8649
XGBoost	6,653	1.5412	3.5458	0.8381
CNN-LSTM	6,653	2.0146	3.6451	0.8289

06

DEMO



**Thanks for
your attention!**

